
TolTEC Raw-Timestream Conditioning

Learn–Resolve–Apply Filtering, Response,
and Scientific Consequences

Science-Team Rationale

Scientific contract version	v0.1
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Method	Implementation-blind scientific authorship

Scope of this revision. It defines why RTC operations are selected, what is learned, what becomes immutable, how astronomical modes are changed, and which claims remain available afterward. It does not claim implementation conformity, representation fidelity, observational performance, science qualification, validation, or production readiness.

Executive abstract

Raw-timestream conditioning is a scientific method, not merely a sequence of array operations. Despiking decides which measurements are inconsistent with an intended signal model. Donor replacement redistributes information between detectors or times. Notches and broad-band filters remove temporal modes and alter amplitude, phase, support, and covariance. Decimation changes temporal identity and can alias rejected modes into the retained band. Each choice can therefore change a point-source peak, effective PSF, centroid, extended-source response, calibration transfer, or scientific eligibility without changing a unit label.

SCI-RTC v0.1/r0.4 organizes those choices as **learn–resolve–apply**. Learn uses a declared population and diagnostics to estimate candidate parameters and uncertainty. Resolve combines that evidence with fixed policy and admits one complete immutable RTC plan. Apply executes exactly that plan and does not relearn or retune. An online adaptive filter is a different, time-varying estimator and remains unavailable unless separately authorized. Missing evidence or tolerances are not passes.

Each resolved response is evaluated against the astronomical temporal spectrum induced by the realized scan velocity, projected beam, source class, cadence, and preceding response. A fixed cutoff is not equally safe for all scan speeds or directions. A notch that removes an interference line also removes whatever astronomical response lies in its width. A detector-static `flxscale` cannot restore removed temporal modes, and calibration transfer across different RTC plans requires explicit response accounting or observational evidence.

The output is an atomic bundle of signal, exact plan identity, response, timing convention, support, influence, validity, uncertainty status, and provenance. RTC signal remains raw detector $\Delta f/f$ in every role; absolute `flxscale` and target atmosphere belong to a distinct downstream SCI-CAL product. When a required predicate is unresolved, the affected operation or claim is explicitly unavailable. The formal appendices are the sole normative authority; Sections 1–12 explain their scientific meaning.

Reading rule. Narrative text explains the shared authority but cannot weaken a requirement, choose a missing numerical value, or convert an unavailable state into a pass.

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1 Executive summary and present status

RTC is the scientific transformation between an admitted aligned detector stream and a conditioned stream whose temporal and detector-mixing response is explicit. Its purpose is not to make every sample finite or every spectrum smooth. Its purpose is to address declared contamination and sampling problems while preserving a quantitatively adequate response for the intended astronomical estimator.

The v0.1 scientific method is learned-then-fixed. Evidence may be learned from a declared population, but application begins only after resolve has produced one complete immutable plan. Online adaptation is a different estimator and is unavailable in this revision. Fixed plans use the same resolve and apply discipline even when no parameter is learned from the observation.

Every operation is interpreted through ten questions: purpose, signal model, operator, design parameters, learning, resolution and application, astronomical consequences, calibration consequences, validation, and unavailable states. Those questions are applied below to despiking, donors, masks, notches, broad-band and general filters, state, boundaries, non-finite handling, and decimation.

The current document is a draft for scientific-owner review. It establishes no production numbers. Open decisions remain open in the ledger; their dependent operations are unavailable. The formal core defines algebraic and semantic authority only. Implementation conformity, representation fidelity, observational performance, science qualification, and production readiness are explicitly unassessed.

Present decision boundary. The owner is being asked to approve the scientific architecture and later resolve the enumerated numerical and policy choices—not to infer those choices from current software or historical practice.

Role-specific RTC plans. The roles share one raw signal boundary and one operator order, but they do not inherit one another’s learning population, plan, interval, cadence, or consumer. “Selected” below means owner-resolved for that role; unresolved entries remain unavailable rather than inheriting a software default.

Role	Learning population			Ordered RTC operations	Parameters frozen
Beammap	Exact	designated	Beammap scans	ALIGN parent → selected despike/donor/mask → ordered filters → phase-zero selection	Complete Beammap RTC plan
Pointing	Exact	designated	Pointing scans	Same operator classes, with a separately selected Pointing plan	Complete Pointing RTC plan
OOF	Exact	designated	OOF scans	Same operator classes, with a separately selected OOF plan	Complete OOF RTC plan
Science	Exact	selected	science observation or scan set	Same operator classes, with a separately selected Science plan	Complete Science RTC plan
Diagnostic	Exact	named parent and diagnostic sample set		Explicit selected subset; no science-role inheritance	Complete diagnostic-only plan

Role	Apply interval		Output cadence	Response-sensitive consumer
Beammap	Named	half-open Beammap interval	Realized raw RTC grid	SCI-BEAM; later SCI-CAL handoff
Pointing	Named	half-open Pointing interval	Realized raw RTC grid	Pointing/SRC and AST consumers
OOF	Named	half-open OOF interval	Realized raw RTC grid	OOF/SRC response inference
Science	Named	half-open science interval	Realized raw RTC grid	PTC, MAP, FLT, and optional later SCI-CAL
Diagnostic	Exact	declared interval	Declared raw diagnostic grid	Named diagnostic consumer only

2 The learn–resolve–apply paradigm

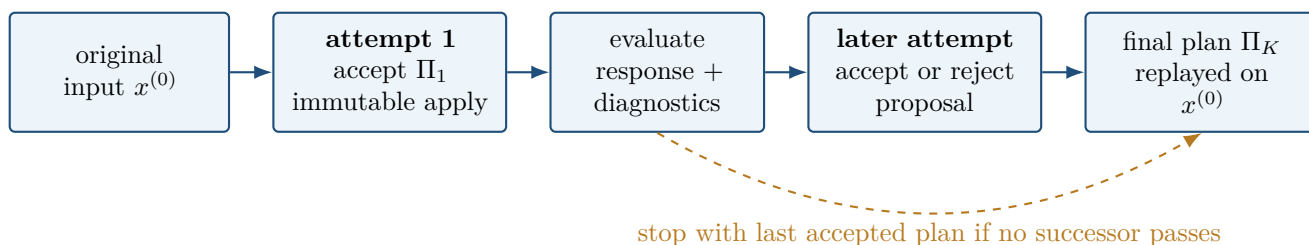
RTC begins with a scientific question: which contaminant or sampling problem must be addressed while retaining adequate astronomical response? The answer is not an operation name. It is a lifecycle that identifies evidence, converts it into a fully specified plan, and applies that plan without hidden adaptation.

Learn. Learn uses a declared population: a separate training interval, a designated part of the same observation, a collection of scans, an observation aggregate, or another named population. It may estimate spike statistics, line centers and widths, filter bands, tap candidates, masks, scan classes, decimation candidates, donor groupings, quality, and uncertainty. Each diagnostic carries estimator, unit, support, selection, and failure state. Learning from the observation later conditioned makes the operator data-dependent. Conditional linear response at fixed state does not eliminate selection bias or parameter-estimation uncertainty.

Resolve. Resolve combines evidence with owner-approved policy. It accepts or rejects every scientific predicate and produces one immutable plan containing sequence, coefficient identity, rates and conventions, order, bands and tolerances, notch state, applicability, masks, donor policy, edges, guards, resets, decimation phase, output timing, expected response/support, uncertainty, and downstream compatibility. Missing evidence or tolerance leaves the dependent plan unresolved.

Apply. Apply consumes exactly one resolved plan. It does not move a notch, change a cutoff or tap count, switch donor policy, change decimation, silently fall back, or refit a parameter. Restart restores the same plan and rejects incompatible inputs. Realized state records what ran without rewriting the request or resolved plan.

V0.1 also permits **bounded iterative notch-plan refinement**. This is an outer finite sequence of attempts over complete accepted plans. Parameters may change only after one apply has finished and a successor attempt has admitted a new complete plan. A rejected or stop attempt does not advance the accepted-plan index or manufacture an evaluation product. Each apply remains immutable. This is the v0.1 scientific meaning when “adaptive notch” is used; it is not continuous coefficient tracking and is not the online-adaptive estimator of SCI-RTC-DEF-025.



Attempt a learns from an exact population and the evaluation product of the current accepted plan Π_{k_a} . Resolve returns a complete cumulative proposal $\tilde{\Pi}_a$ and a typed accepted, rejected, or stop disposition. Only acceptance advances the plan index and creates a new evaluation product. The initial evaluation is explicitly $x^{\text{eval},(0)} = \mathcal{A}_{\Pi_0}(x^{(0)})$, including when Π_0 is a nonidentity fixed plan. A successor proposal names all retained, modified, removed, or rejected notches and every broad-band, state, mask, edge, timing, and decimation choice. It is never merely “add this notch to whatever ran before.” The final production product normally replays Π_K against the original input, avoiding unintended repeated rounding, padding, state transients, duplicate filtering, stale masks, or ambiguous parentage.

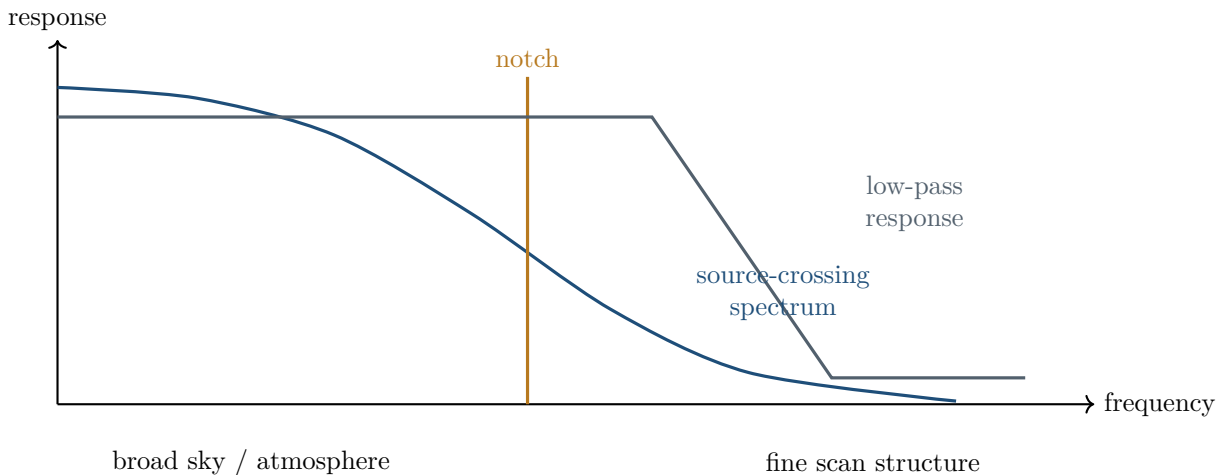
A sequential cascade from the preceding filtered product is a different scientific operator. It is allowed only when selected explicitly and when every immutable intermediate parent, response, state, phase, support, covariance, and uncertainty is retained. Learned-then-fixed and online-adaptive filtering remain different: in the latter, parameters evolve inside one apply pass. V0.1 does not authorize that branch.

Lifecycle test. For each attempt identify its attempt index, current accepted-plan index, learning and parent products, complete proposal, predicates, and disposition. For each accepted plan show that apply did not modify it. Then identify the finite stop cause and the final selected plan. If any link is absent, the refinement claim is unavailable.

3 Temporal frequencies, scan speed, and astronomical spatial scales

The same temporal frequency represents different angular structure in different scans. For a locally Gaussian effective beam with covariance Σ and angular scan velocity $\mathbf{v}$, the characteristic crossing time is equation (SCI-RTC-EQ-023). It uses the beam projected along the actual scan direction, not merely a scalar array FWHM. Source extent, detector and telescope response, prior RTC stages, and cadence further shape the exact source-times-response spectrum.

High frequencies correspond to finer scan-direction structure. Low frequencies contain broad sky structure together with baselines, atmosphere, and slow instrumental behavior. These do not cleanly separate source and contaminant. A Gaussian is not band-limited; extended emission can share low frequencies with atmosphere; a coherent line can overlap a source crossing. The shorthand $f \sim v/\theta$ is pedagogical only.



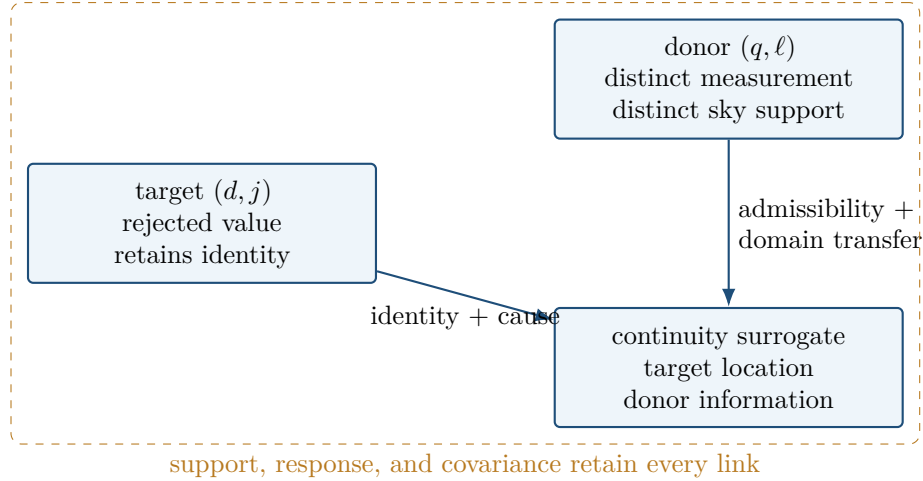
Changing speed moves astronomical modes relative to a fixed cutoff or notch; changing direction changes projected beam width. Every filter is assessed over declared speed, direction, beam, source extent, and source class using the exact realized response. Preserved dimensions or units do not establish preserved peak, PSF, centroid, extended response, or absolute mode.

4 Despiking and donor replacement

Despike purpose and model. Despiking identifies samples inconsistent with an intended detector-plus-sky signal under a stated contaminant model. The statistic, baseline, threshold, support, mask, and false-positive control define the method. Learn may estimate statistics from a declared population; resolve freezes the estimator, threshold, support, boundaries and disposition; apply evaluates them unchanged. A false positive removes sky signal, while a false negative can ring through a long filter. Convergence of an optimizer is not evidence of unbiased retained signal.

A **donor** is an admitted detector-time measurement used to construct a continuity surrogate for a rejected target cell. The output remains located at the target detector and time, but its numerical information comes partly or wholly from the donor. This can prevent a spike, NaN, or gap from contaminating a later filter; it is not automatically reconstruction of the missing astronomical measurement.

Donor and target can differ in focal-plane position, instantaneous sky, beam, atmosphere coupling, baseline, electronic response, loading, noise, and covariance. Numerical convention transfer, donor admissibility, and scientific eligibility are distinct questions. Under $\chi_i = \phi_i x_i$, equation (SCI-RTC-EQ-002) uses ϕ_q/ϕ_d only for exact compatible occurrences with nonzero target factor. The ratio solves a raw-domain convention problem; it does not prove that both detectors observed the same sky. Beammap donor factors must come from an identified prior APT or other pre-existing authority, never the new **flxscale** being derived from that same Beammap.



Resolve selects topology, time relation, masks, quality, scoring, ties, weights, baseline treatment, reuse and fallback. Apply cannot change them. Reused donors create correlations and no exposure. A directly selected ALIGN-synthesized or RTC-replaced occurrence is universally excluded as an independent detector measurement. Noncenter influence through a filter or other support remains visible with its cause and numerical dependency; the declared downstream consumer, not RTC, determines its eligibility. Validation injects spikes, invalid samples, factor differences, sky mismatch, donor reuse, source crossings and mask-edge cases, measuring bias, covariance, affected support and data loss.

5 Notch filtering and learned interference removal

A notch suppresses a narrowband coherent disturbance while preserving as much surrounding astronomical information as possible. A line’s physical origin is not inferred from frequency or coherence. The scientific question is what astronomical transfer, phase, support and covariance remain, not only whether the line disappeared.

A resolved notch declares center, rate and frequency convention, width or Q, depth, family and order, poles/zeros where applicable, phase and group delay, state, transient and guard, applicability, learned uncertainty, and drift tolerance. Equation (SCI-RTC-EQ-024) is a generic pole-zero form; the exact realized transfer is authoritative. A too-narrow notch leaves a drifting or finite-width line; a broad notch removes more sky and noise. High Q usually means longer memory. Greater depth can increase numerical sensitivity. Forward/backward use changes magnitude, order, phase and boundaries. Overlapping notches require combined-response analysis.

Learn defines detection, parameter estimation, significance, false-detection control, and the declared detector, network, array, and scan scope. Resolve freezes a common plan and its drift-validity domain. Apply uses it unchanged. A line not found in learn does not prove absence in apply; fitting noise can remove real sky modes. An evolving notch is an online-adaptive estimator with separate update, response, stability and restart authority; it remains unavailable in v0.1.

Successor-attempt assessment. A later learn pass first asks whether the accepted plan had its intended consequences: line attenuation and residual significance, center error or drift, adequate width/depth, ringing, transient, stability, edge loss, phase and group delay, point-source amplitude and centroid, effective PSF, extended-source transfer, covariance, and support. On a declared fixed-state LTI interior, the predicted filtered PSD may be compared with $P_{\text{pred},k}(f) = |H_k(f)|^2 P_0(f)$. That diagnostic is not universal at masks, edges, state transitions, selections, or nonstationary intervals. The spectral hole made by a notch is expected filter response, not new evidence that the original disturbance was absent.

The second question is whether conditioned diagnostics warrant a successor notch. A peak is not sufficient. Resolve compares original, predicted, and conditioned PSDs; cumulative response; line uncertainty; astronomical temporal spectrum; transfer and alias budgets; and detector/network/array/scan/interval consistency. It rejects notch-edge artifacts, ringing, transients, leakage, normalization changes, numerical features, and structure where response is too small for inference. Held-out scans are preferred where practical. Reusing the same data for selection and assessment is labeled observation-internal, data-dependent consistency, not independent validation.

Bounded admission and stopping. A successor is admitted only when its line and parameters are identifiable, it improves the interference metric, and cumulative astronomical amplitude, centroid, PSF, phase, registration, ringing, stability, support, cadence, response, uncertainty, and provenance budgets all pass. Missing evidence or tolerance is not a pass. Rejection leaves the last accepted plan available and preserves the proposal and cause. Iteration stops under an owner-approved finite policy governing attempt count, notch count, removed bandwidth, scientific loss, delay, ringing, edge loss, minimum improvement, plan stability, false-line control, and fallback. Maximum-attempt termination is not convergence. Alternating line sets or unstable parameters are explicitly nonconvergent.

Example. Attempt 1 learns a strong coherent line and accepts a one-notch plan Π_1 . After immutable apply, Attempt 2 learns from its diagnostic product and reveals a weaker line. It accepts a complete two-notch plan, which is replayed once against the original input. In the counterexample, a peak at the first notch edge agrees with predicted ringing or response artifact; the candidate is rejected and the one-notch plan remains final. More notches are not automatically better.

`flxscale` cannot restore removed frequencies. Beammap and science observations with different notch plans do not automatically share point-source response, even under the same label. Validation varies center error, drift, width, depth, Q/order, state, direction and overlap, measuring residual line, source transfer, ringing, transient, amplitude, centroid, covariance and edge loss.

6 Low-pass, high-pass, and band-pass filtering

Low-pass. A low-pass suppresses frequencies above a declared band. It can reduce high-frequency noise, prepare for decimation, control aliasing, or define a retained band. It can also attenuate a fast source crossing, broaden or shift the PSF, reduce peak, alter morphology, ring, increase covariance, and reduce usable edge support. Its plan states passband and stopband edges, transition width, ripple, attenuation, phase/delay, family, taps/order, precision, edges and guards.

High-pass. A high-pass can suppress drift or broad atmosphere, but also removes large angular scales, extended flux, absolute or slow sky modes, beam wings and baselines. It can generate negative lobes around compact sources. Any admitted branch needs an explicit recoverable-mode statement.

Band-pass. A band-pass defines a finite scan-dependent temporal and spatial band. Both rejected ends require justification; its combined response, not two cutoff labels, is authoritative.

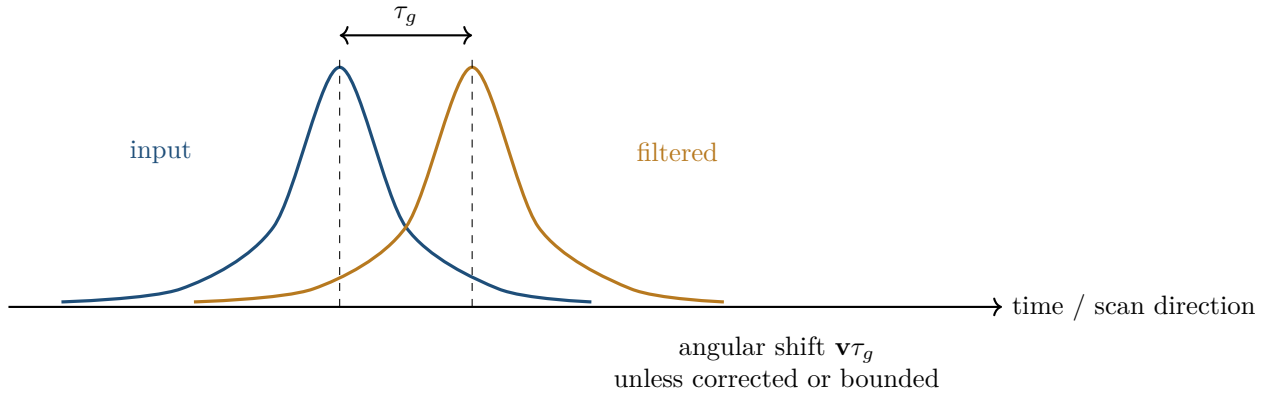
V0.1 admits these classes only through a resolved selected policy and chooses no universal cutoff or family. Where no role is named, the branch is unavailable. Studies vary edges, ripple, attenuation, family, precision, speed, projected beam and source extent, measuring point and extended response, ringing, delay, covariance and usable fraction.

7 Filter order, tap count, phase, and temporal registration

FIR tap count is a scientific design result, not an arbitrary integer. Equation (SCI-RTC-EQ-025) chooses the smallest order satisfying passband, stopband, phase, alias, edge and realizability requirements. A window estimate can initialize a search; only exact coefficients and response close the predicates.

More taps can sharpen transition or attenuation, but increase computation, support, covariance footprint, affected outputs after replacement, and edge loss. They can ring around sources and spikes. Quantization can degrade ripple, attenuation, notch placement and stability. A causal linear-phase FIR has equation (SCI-RTC-EQ-026); forward/backward use can remove ideal interior phase while squaring magnitude, doubling effective order and changing boundaries.

Delay creates scan-direction displacement by equation (SCI-RTC-EQ-027). The output grid time can be simple while response is centered elsewhere. Beammap and Pointing are especially sensitive. Resolve requires zero phase, a shifted time, coordinate correction, response-aware estimator, or a bound showing residual position and PSF effect is negligible.



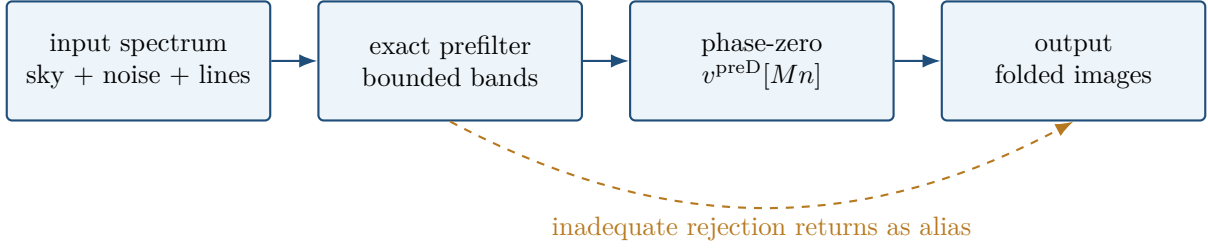
Validation varies taps, family, precision, bands, speed, beam and source class, measuring amplitude, shape, delay, ringing, aliasing, covariance, coordinate registration and usable length. Coefficient algebra alone is not science qualification.

8 Decimation and alias control

V0.1 phase-zero downsampling, equation (SCI-RTC-EQ-011), is point selection $v^{\text{preD}}[Mn]$, not averaging and not anti-alias filtering. It changes software cadence, Nyquist frequency, temporal identity, support, covariance, aliasing, and coordinate registration. The preceding filter determines the rejected-band power that remains to fold.

Equation (SCI-RTC-EQ-014) shows that components separated by integer multiples of output rate can appear at the same output frequency. Before resolving M , the exact prefilter must meet approved tolerances in all folded bands and the declared source domain. A Gaussian is not band-limited, so “safe” means bounded residual transfer and alias, not exact preservation.

Factor choice uses input cadence, exact prefilter, bands, smallest projected beam-crossing time, maximum valid scan speed, source class, required output cadence, delay/coordinate accuracy and consumer compatibility.



Selected grid time t_{Mn} does not erase preceding delay or broad support. Science may require zero phase, shifted time, coordinate correction, response-aware estimation, or a negligibility bound. Tests include on/off-bin tones, varied scan speeds/directions, smallest-beam sources, delay and coordinate checks, aliases, incomplete strides, edges and resets.

9 Masks, edges, state, and non-finite handling

Masks. A source mask protects a coordinate region or changes data used to learn an operation. It requires exact frame, topology, binding, support, validity, geometry, dilation and boundary. It is not acquisition validity or confidence. It can protect real sources while allowing spikes inside the region, and can change normalization, create discontinuities or ringing, and alter state. Learn, resolve and apply state whether the mask is learned, fixed or consumed and what happens if coordinates are unavailable.

FIR boundaries. Complete-context rejection, zero padding, reflection, extension and local renormalization are different operators. A guard removes outputs; it does not prove a residual IIR tail negligible without a criterion. Short and empty scans need explicit dispositions.

IIR state and non-finite values. A plan states initialization, reset/carry boundaries, stability, transient, gaps and cross-scan or cross-observation behavior. A non-finite input can poison state until declared reset or recovery and cannot silently become zero. Learn may estimate state or settling; resolve freezes the convention; apply records actual boundaries and rejects incompatible continuation. Tests place impulses, sources, masks and NaNs at every boundary and split/rejoin streams at state boundaries.

10 Calibration, Beammap response, and downstream interpretation

RTC distinguishes detector-static `flxscale`, sample-dependent target atmosphere correction, detector mixing, single-detector temporal filters, and data-dependent learned or adaptive operators. Their order is not chosen by convenience.

The scientific owner selected one v0.1 cross-package order: RTC remains raw $\Delta f/f$ through compatible donor replacement, temporal filtering, and phase-zero sampling. A valid `flxscale` ratio may transfer donor convention inside raw replacement, but that ratio is not absolute calibration. Only after the complete raw RTC bundle exists may SCI-CAL apply selected absolute `flxscale` and sample-dependent target-atmosphere correction. Equation (SCI-RTC-EQ-004) records this handoff; no calibrated RTC branch or alternative interleaving is authorized.

Filtering can preserve units while changing peak, PSF, extended response, phase, centroid and covariance. Calibration does not restore removed modes. A Beammap-derived `flxscale` belongs to its complete RTC lineage. A target with a different RTC plan can have a different point-source transfer. Names such as “low-pass” or “notch” are not compatibility: coefficients, rate, phase, state, masks, edges, decimation and response must agree or be accounted for.

Cross-plan calibration needs response calculation, declared renormalization, or observational evidence over the intended source and scan domain. The RTC bundle preserves the plan needed to compare calibrator and target. Without compatibility, the unit may remain meaningful while the photometric or PSF claim is unavailable.

Beammap circularity guard. New `flxscale` derived from a Beammap cannot be used as donor scaling inside that same Beammap RTC lineage; donor factors come from explicit pre-existing authority.

11 Covariance, support, and scientific eligibility

Output dependence expands through alignment, donors, taps or state, boundaries and sampling. Support—not storage shape—governs information flow. Reused donors correlate targets, overlapping FIRs correlate neighbors, an IIR links to its state boundary, and decimation erases none of those facts.

At fixed state, equations (SCI-RTC-EQ-016a) and (SCI-RTC-EQ-016b) propagate an admitted mean and covariance through the same operator. Data-dependent selection or learned parameters add selection variation and possibly cross terms. Conditioning does not set them to zero. Unknown covariance is unavailable, not diagonal, white or zero.

Eligibility differs from computability. V0.1 universally excludes a directly selected ALIGN-synthesized or RTC-replaced occurrence as an independent detector measurement. For synthesis or replacement elsewhere in full support, RTC preserves value, response, cause, linkage, and numerical influence but does not impose universal ineligibility. Each downstream consumer declares how those noncenter influences affect its estimator. Consumers cannot erase causes or strengthen unavailable response. Donor studies must measure bias, influence extent, and the data loss produced by each consumer policy.

Response, uncertainty and eligibility accompany signal in the atomic bundle of equation (SCI-RTC-EQ-022). Finite TOD without exact plan, support, response or typed unavailability, influence and validity is incomplete.

12 Validation and open owner decisions

Algebraic fixtures are necessary but not sufficient. Validation varies the scientific design choices and measures the observables they can change.

Study	Variations	Measurements
Notch	center error, drift, width, depth, Q/order, state, direction, overlap	residual line, sky transfer, ringing, transient, amplitude, centroid, covariance, edge loss
FIR/bands	taps, edges, ripple, attenuation, family, precision, speed, beam, source extent	peak/shape, extended transfer, delay, ringing, alias, covariance, usable fraction
Donor	spikes, invalids, factors, sky mismatch, noise, reuse, sources, mask edges	stabilization, bias, covariance, influence extent, scientific data loss
Decimation	on/off-bin tones, speeds/directions, smallest beam, delay, aliases, strides, resets	retained/aliased transfer, timing, coordinate shift, PSF, support, edges

Each study preregisters domain, estimator, tolerance and falsifier. Passing coefficient algebra does not establish science qualification. Observational performance, science impact and production readiness are separate layers.

The owner ledger retains unresolved authority for learning populations and freeze points; despikedonor/mask policies; notch and broad-band purpose and design; filter families, order/taps, tolerances and precision; edges, state and non-finites; decimation and alias bounds; phase/coordinate registration; CAL/RTC compatibility; diagnostics; uncertainty; and bundle members. Online adaptation remains deferred. An open entry blocks only its named operation or claim and never authorizes a default or silent fallback. Cross-package follow-up routes CAL, BEAM, AST, PTC, VAL, MAP and filtering questions without letting RTC assume their answers.

Success criterion. For every operation, a scientist can explain purpose, learned evidence, resolved numerical choice, immutable application, changed astronomical modes, response and calibration effects, validation evidence, and remaining unavailable claims.

A Shared normative authority and notation

Normative core stamp: SCI-RTC-v0.1/r0.4. The six files in `src/common/` form one authority. Text outside this shared core is explanatory or conformance guidance and cannot alter a definition, equation, assumption, requirement, prediction, unavailable state, or owner decision.

SCI-RTC begins with an admitted aligned raw primary detector stream and ends with an atomic, role-specific raw conditioning bundle. ALIGN, AST, and SCI-BEAM retain authority for the upstream meanings they produce; SCI-CAL is the downstream authority that may calibrate an exact RTC bundle. PTC, VAL, MAP, BEAM, NOI, MAP-003, FLT, SRC/MODE, and FRUIT retain authority for their downstream science. This contract owns the realized conditioning order and its scientific consequences; it does not infer current software behavior.

The words **shall** and **shall not** identify requirements. **Required unavailable** means that the affected product or claim shall not be reported as complete. **Optional unavailable** is a typed, scientifically inert state. Missing, disabled, automatic, rejected, invalid, failed, unavailable, unsupported, and omitted remain distinct states.

A.1 Indices, sets, and role labels

Symbol	Normative meaning
o	Immutable observation identity.
$d, q \in \mathcal{D}_o$	Exact admitted detector occurrences; d is a target and q may be a donor. A detector index or row number alone is not this identity.
r	Native acquisition row; it is not an aligned or conditioned slot.
$j \in \{0, \dots, N - 1\}$	Ordered aligned input slot on the admitted ALIGN grid.
$n \in \{0, \dots, N_y - 1\}$	Ordered phase-zero RTC output slot.
$\mathcal{R}$	RTC product role, including Pointing, OOF, Science, Beammap, diagnostic-only, or another separately approved role; role changes policy and consumers, not the raw signal identity.
x_{dj}^A	Admitted aligned primary xs value: raw, uncalibrated detector $\Delta f/f$ with declared sign, baseline/reference, support, and validity.
χ_{dj}	Detector-static flxscale comparison coordinate used only to define compatible donor transfer; it is not a calibrated RTC product.
u_{dj}	Raw $\Delta f/f$ value immediately before the ordered RTC filter chain.
v_{dj}^{preD}	Final raw $\Delta f/f$ stream after every admitted RTC replacement/filter stage and immediately before phase-zero point selection.
y_{dn}^{RTC}	Conditioned raw $\Delta f/f$ RTC output at the selected phase-zero slot.
c_{dn}^{CAL}	Distinct downstream SCI-CAL output, when authorized: top-of-atmosphere, point-source-equivalent mJy per fixed nominal beam.
ϕ_d	flxscale for one exact detector occurrence under the declared multiplicative convention and calibration domain.
A_α	Imported, realized ALIGN mapping with state α ; its scientific meaning is not owned by RTC.
C_κ^{CAL}	Downstream SCI-CAL operator with state κ , including selected absolute flxscale and target-atmosphere correction; RTC does not apply it.
B_ω	Realized despike/replacement operator conditioned on selection state ω .
F_{k, θ_k}	Realized ordered FIR, IIR, notch, or admitted mask-controlled stage k .
$D_{M,0}$	Integer phase-zero point selector of factor $M \geq 1$.

Symbol	Normative meaning (continued)
Ω	Complete realized conditioning state, including all upstream bindings and every selected stage, exception, state, edge rule, and output interval.
$\mathcal{L}$	Declared learning population with exact identity, selection, support, masks, exclusions, and compatibility domain.
Λ	Learned evidence and uncertainty produced by declared diagnostics; it is not authorization to apply.
Π	One immutable resolved RTC plan admitted from learned evidence and fixed scientific policy.
$\mathcal{A}_\Pi$	Apply operator that consumes plan Π without changing it.
$a \in \{1, \dots, N_A\}$	Refinement-attempt index and finite owner-bounded maximum attempt count.
$k \in \{0, \dots, K\}$	Accepted-plan index; $K \leq N_A$ is the final accepted-plan index.
k_a	Accepted-plan index available at the start of attempt a .
$\tilde{\Pi}_a$	Complete cumulative successor plan proposed by attempt a ; it is not an accepted plan unless its disposition is accepted.
δ_a	Typed attempt disposition in $\{\text{accepted}, \text{rejected}, \text{stop}\}$.
Π_k	Complete cumulative accepted plan k ; Π_0 is the admitted initial fixed or identity plan.
$x^{\text{eval},(k)}$	Exploratory evaluation product for accepted plan k , normally obtained by applying Π_k to original admitted input $x^{(0)}$.
$K_{dn,qj}$	Local detector-time response of one output cell to one acquired input cell.
$\mathcal{S}_{dn}$	Exact transitive input support of output (d, n) , including alignment, donors, filters, state, and selection.
$\mathcal{I}_{dn}$	Cause-preserving transitive influence set for output (d, n) .

A.2 Identity and interval conventions

Observation, Tune, array, network/interface, detector occurrence, selected APT row/binding, native row, aligned slot, scan, science interval, context interval, conditioned output slot, stage, and product identities are non-interchangeable. Array identity is one of **a1100**, **a1400**, and **a2000**; array ID, array index, network ID, detector index, and container position do not substitute for one another. Cross-artifact identity requires an explicit occurrence- and product-scoped binding. Equal row numbers do not establish identity.

Primary detector timestreams are samples on rows and detectors on columns unless a product explicitly declares another ordered shape. Every interval is half-open. Science windows, physical scans, processing chunks, filter context, output selections, and learned-plan scan sets retain separate identities. Native, aligned, and output indices never silently renumber one another.

Time is seconds on the admitted ALIGN-assigned grid. The approximately 8-ms rate relationships and the $0.5\times$, $2\times$, and $4\times$ family are conditional software-grid facts, not proof of a detector integration event, absolute epoch, or astrometric correction. Coordinate-dependent Pointing, OOF, and Beammap controls use a declared AltAz tangent plane; Science controls use a declared equatorial J2000 TAN relation. RTC performs no implicit frame conversion.

A.3 Logical and availability notation

$\mathbb{I}[\cdot]$ denotes an indicator, $\mathbb{E}$ expectation, Cov covariance, and $\text{supp}(\cdot)$ operator support. The logical symbols $\wedge$, $\vee$, and $\neg$ have their usual meanings. A quantity decorated available is present with its required identity, domain, and validity; a quantity decorated unavailable is explicitly unavailable. Neither zero nor NaN is an availability encoding.

End of notation; shared definitions follow without a change of authority.

B Normative definitions

SCI-RTC-DEF-001 – Admitted aligned stream.

An immutable observation-scoped primary **xs** array whose physical observable is raw, uncalibrated detector $\Delta f/f$, with sign, zero/reference, baseline, ordered detector binding, assigned sample grid, cadence, phase, support, origin/synthesis state, validity, scan partition, and mapping lineage are supplied by the responsible upstream authority. Admission does not upgrade unavailable physical event timing.

SCI-RTC-DEF-002 – RTC product-role contract.

The tuple $(\mathcal{R}, s_{\mathcal{R}}, b_{\mathcal{R}}, \mathcal{V}_{\mathcal{R}}, \mathcal{C}_{\mathcal{R}})$ naming role, sign, reference/baseline, valid domain, and required consumers. Every RTC role retains raw $\Delta f/f$ signal identity. A calibrated downstream detector signal is a distinct SCI-CAL product c^{CAL} in the top-of-atmosphere, point-source-equivalent mJy-per-fixed-nominal-beam convention; it is not another meaning of **xs**.

SCI-RTC-DEF-003 – Exact detector occurrence.

An observation-local layered binding containing observation, Tune, network/interface, network-local tone or admitted row slot, selected APT identity/digest, and row association. A transient column number is only a locator.

SCI-RTC-DEF-004 – Downstream CAL handoff.

An immutable RTC-to-SCI-CAL boundary carrying the raw conditioned parent, detector binding, complete RTC plan/response/support, selected **flxscale** authority, target-atmosphere lineage, validity, uncertainty availability, and provenance required by SCI-CAL. Absolute **flxscale** and target atmosphere are applied only after RTC; RTC does not derive, apply, repair, or promote the calibrated product.

SCI-RTC-DEF-005 – Compatible flxscale pair.

Two finite factors ϕ_q and ϕ_d that are valid for the exact donor and target detector occurrences under the same multiplicative calibration convention, quantity domain, reference plane, and applicable support, with $\phi_d \neq 0$. Compatibility is not established by matching row numbers or finite values alone.

SCI-RTC-DEF-006 – Selected conditioning policy.

A versioned, admitted specification of despike detection, donor eligibility, donor scoring and ties, replacement and fallback, source protection, filter stages and order, exact coefficients and normalization, numeric precision, state/reset, boundaries/edges, non-finite handling, sampling mode, phase-zero factor, flags, influence aggregation, diagnostics, and failure behavior. It is selected authority, not an implementation default.

SCI-RTC-DEF-007 – One-way state record.

The ordered record $(\Theta^{\text{req}}, \Theta^{\text{eff}}, \Theta^{\text{obs}}, \Theta^{\text{learned}}, \Theta^{\text{resolved}}, \Theta^{\text{real}})$. Requested state preserves the accepted request; effective state is the admitted typed plan; observation-resolved state binds exact observation facts; learned state records evidence when learning occurs; resolved state is the immutable admitted plan for every fixed or learned operation; and realized state records what ran. A later state never rewrites an earlier state.

SCI-RTC-DEF-008 – Realized conditioning operator.

The chronological composition of every admitted alignment, raw-domain replacement, mask, FIR/IIR/notch, boundary, state, non-finite, and phase-zero sampling action, including all affine state terms and exception intervals. Annotations that affect values, support, influence, or availability are part of the operator.

SCI-RTC-DEF-009 – Phase-zero sampling.

Point selection of the pre-sampling stream at input indices $j = Mn$ for one integer factor $M \geq 1$. It is not an arithmetic mean, block aggregate, or implicit anti-alias filter.

SCI-RTC-DEF-010 – Complete realized response.

An exact local or factorized representation that includes every enabled response-changing stage, detector mixing, mask, state, edge rule, exception, and sampling phase on its declared support. If such a representation is not truthfully available, response state is unavailable; a partial kernel is not complete response.

SCI-RTC-DEF-011 – Direct source cell.

An aligned cell that enters an RTC value before transitive propagation. It may be original, ALIGN-synthesized, or RTC-replaced. Numerical participation does not confer independent scientific exposure.

SCI-RTC-DEF-012 – Transitive influence.

The cause-preserving closure from every original, synthesized, replaced, masked, edge, state, and non-finite source cell through all nonzero realized alignment, donor, filter, state, and sampling dependencies to an output. Influence is broader than the selected point and may cross detector columns.

SCI-RTC-DEF-013 – Direct exclusion and consumer eligibility.

An output whose directly selected source occurrence is ALIGN-synthesized or RTC-replaced is universally excluded as an independent detector measurement, even when finite. Synthesis or replacement elsewhere in its

transitive support remains a mandatory cause-preserving influence input, not a universal RTC eligibility decision. Each downstream consumer owns the declared policy that maps that influence, response, and support to its own eligibility.

SCI-RTC-DEF-014 – Coordinate-dependent control.

A source mask or response operation that consumes an AST coordinate identity, frame, topology, detector binding, validity, uncertainty where required, and support. Invalid or unavailable coordinates are not outside-source values.

SCI-RTC-DEF-015 – Fixed sampling mode.

A request whose admitted factor, realizable prefilter, rate, phase-zero rule, and downstream compatibility are resolved without learning them from the observation.

SCI-RTC-DEF-016 – Learned sampling mode.

An optional request that evaluates an admitted finite candidate set and resolves one immutable observation-wide cadence/filter plan before apply. Its admission uses the smallest admitted beam, maximum valid in-scan speed, analytical beam-times-realized-filter response, exact phase-zero alias accounting, sampling sufficiency, filter realizability, and downstream compatibility. Noise-aware, per-array, per-scan, heterogeneous-cadence, and continuous adaptation are outside v0.1.

SCI-RTC-DEF-017 – Conditional statistical covariance.

The covariance obtained from an admitted input covariance through a fixed realized operator while holding selection and nuisance state fixed. It is not total uncertainty and does not imply a diagonal, white, stationary, or independent noise model.

SCI-RTC-DEF-018 – Atomic RTC bundle.

One role-specific completion unit containing conditioned raw $\Delta f/f$ TOD; ordered detector/sample/scan/support and parent identity; complete response or typed unavailability; cause-preserving flags and influence; separated validity and eligibility inputs; conditional uncertainty/covariance status; one-way state and provenance; required diagnostics; downstream CAL handoff state; and completion/failure state. TOD alone is not this bundle.

SCI-RTC-DEF-019 – Scientifically named diagnostic.

A quantity with an explicit estimator, unit, support, selection/mask, validity, uncertainty/availability, and policy role. It is either inert, advisory, or an input to an explicitly selected policy; those roles are not interchangeable.

SCI-RTC-DEF-020 – Claim layer.

One of algebraic contract correctness, implementation conformity, representation fidelity, observational performance, science-impact qualification, or production readiness. Evidence at one layer does not establish another.

SCI-RTC-DEF-021 – Declared learning population.

The exact training interval, designated same-observation interval, scan collection, observation aggregate, or other named population used to infer an RTC quantity, including identity, support, selection, masks, exclusions, signal role, compatibility domain, and completeness. A population is not defined by whatever samples happen to be available during execution.

SCI-RTC-DEF-022 – Learned evidence.

The output of named diagnostics applied to a declared learning population, including candidate parameters, estimator, unit and convention, support, quality, uncertainty, validity, and failure state. Learned evidence proposes or constrains a plan; it does not authorize apply.

SCI-RTC-DEF-023 – Resolved RTC plan.

One immutable observation- and role-bound specification admitted only after all required scientific predicates pass. It contains operation sequence, family, exact coefficients or identity, rates and conventions, order/taps, bands and tolerances, notch state, applicability, masks, donor policy, edges, guards, resets, decimation and timing, expected response/support, uncertainty, validity, and downstream compatibility.

SCI-RTC-DEF-024 – Apply state.

Execution of exactly one resolved plan on compatible admitted input. Apply records actual intervals, exceptions, resets, outputs, and completion but does not relearn, retune, change policy, or rewrite the resolved plan.

SCI-RTC-DEF-025 – Online-adaptive estimator.

A separate time-varying, data-dependent estimator whose coefficients or other operator parameters evolve during apply under a declared update rule. Its learning support, stability, response, uncertainty, guard, reset, and restart semantics are not those of learned-then-fixed filtering. It is unavailable in v0.1 absent successor authority.

SCI-RTC-DEF-026 – Scientific filter design specification.

The physical purpose, source/contaminant model, admitted scan/beam/source domain, pass/transition/stop or notch constraints, phase and registration, alias bound, support/guard, numeric precision, realizability, validation, and failure criteria that justify a filter family and its numerical parameters. An operation label or coefficient

array alone is not this specification.

SCI-RTC-DEF-027 – Bounded iterative notch-plan refinement.

A finite owner-bounded sequence of refinement attempts. Each attempt learns from a declared population and the current accepted-plan evaluation product, resolves a complete cumulative proposal and typed disposition, and advances the accepted-plan index only when that proposal is accepted. Apply of every accepted plan is immutable; rejected or stop attempts do not manufacture a new accepted plan or evaluation product. This is not the online-adaptive estimator of SCI-RTC-DEF-025.

SCI-RTC-DEF-028 – Complete cumulative successor proposal.

The full plan proposed by attempt a and accepted as Π_{k+1} only after its admission predicates pass, containing every retained, modified, removed, and rejected notch disposition; every broad-band stage; exact order, coefficients, precision, state, masks, edges, guards, intervals, decimation, timing, cumulative response/support, uncertainty, validity, and downstream compatibility. “Add this notch to the prior output” is not a complete successor plan.

SCI-RTC-DEF-029 – Final accepted plan and termination disposition.

The last complete plan whose admission predicates passed, together with one typed finite stop cause. No-successor, residual-satisfied, stable-plan, budget/stability/support/compatibility rejection, insufficient evidence, maximum-attempt, oscillating-plan, nonconvergent-parameter, and owner-cancelled states remain distinct. Maximum-attempt termination is not convergence.

C Normative equations and identities

The equations in this section are the sole displayed mathematical authority of both views. They describe an implementation-independent, state-conditioned operator. They neither select numerical policy nor assert implementation conformity.

C.1 Raw RTC domain, donor transfer, and CAL handoff

For an exact detector occurrence i under the admitted multiplicative `flxscale` convention, define the detector-static comparison coordinate

$$\chi_i = \phi_i x_i, \quad \phi_i = \text{flxscale}_i. \quad (\text{SCI-RTC-EQ-001})$$

Equation (SCI-RTC-EQ-001) neither produces a calibrated RTC product nor assigns a role to legacy `responsivity`; target-atmosphere correction is absent. When q is a raw donor and d is a raw target, the unique multiplicative transfer that represents donor-calibrated amplitude in the target raw convention is

$$x_{d \leftarrow q}^R = a_{d \leftarrow q} x_q, \quad a_{d \leftarrow q} = \frac{\phi_q}{\phi_d}, \quad \chi_d^R = \phi_d x_{d \leftarrow q}^R = \phi_q x_q = \chi_q. \quad (\text{SCI-RTC-EQ-002})$$

This equation is defined only for a compatible `flxscale` pair as defined above. Otherwise this transfer is unavailable. An additive calibration convention, a different unit/reference domain, or an alternative donor authority requires its own complete affine equivalence and is not silently folded into equation (SCI-RTC-EQ-002).

Let $\Phi = \text{diag}(\phi_d)$ be the selected detector-static factor operator and $C_{\kappa, \mathcal{R}}^{\text{atm}}$ the target-atmosphere operator owned by SCI-CAL. The distinct downstream calibration operator is

$$C_{\kappa, \mathcal{R}}^{\text{CAL}} = \Phi C_{\kappa, \mathcal{R}}^{\text{atm}}, \quad \mathbf{c}_{\mathcal{R}}^{\text{CAL}} = C_{\kappa, \mathcal{R}}^{\text{CAL}} \mathbf{y}_{\mathcal{R}}^{\text{RTC}}. \quad (\text{SCI-RTC-EQ-003})$$

Equation (SCI-RTC-EQ-003) is an SCI-CAL handoff identity; its output is not `xs` and it is not an RTC stage. The selected cross-package order is

$$\mathbf{x}^{\text{acq}} \xrightarrow{A_\alpha, B_\omega, F_1 \dots F_K, D_{M,0}} \mathbf{y}_{\mathcal{R}}^{\text{RTC}} \xrightarrow{C_{\kappa, \mathcal{R}}^{\text{CAL}}} \mathbf{c}_{\mathcal{R}}^{\text{CAL}}. \quad (\text{SCI-RTC-EQ-004})$$

RTC shall not move the downstream CAL arrow inside replacement, temporal filtering, or phase-zero sampling. The raw donor ratio in equation (SCI-RTC-EQ-002) is convention transfer within B_ω , not absolute calibration.

C.2 Realized conditioning operator

Conditioned on complete realized state Ω , the ordered raw role-specific operator is

$$\mathbf{y}_{\mathcal{R}}^{\text{RTC}} = D_{M,0} F_{K,\theta_K} \cdots F_{1,\theta_1} B_{\omega,\mathcal{R}} (A_{\alpha} \mathbf{x}^{\text{acq}} + \mathbf{a}_{\alpha}) + \mathbf{b}_{\Omega} . \quad (\text{SCI-RTC-EQ-005})$$

Chronological order is read from right to left. The affine terms include only declared replacement, filter-state, boundary, or other authorized contributions. Masks, coefficients, finite handling, context, exceptions, and state are components of Ω , not annotations outside the operator. Conditional on them,

$$\mathbf{y}_{\mathcal{R}}^{\text{RTC}} = L_{\Omega} \mathbf{x}^{\text{acq}} + \mathbf{b}'_{\Omega}, \quad L_{\Omega} = D_{M,0} F_K \cdots F_1 B_{\omega,\mathcal{R}} A_{\alpha}. \quad (\text{SCI-RTC-EQ-006})$$

This is a conditional affine identity, not an unconditional linear, stationary, unbiased, or Gaussian claim.

A realized replacement at target cell (d, j) may be written

$$u_{dj}^R = (1 - p_{dj}) u_{dj} + p_{dj} \left[\sum_{(q,\ell) \in \mathcal{Q}_{dj}} w_{dj,q\ell} a_{d \leftarrow q,j\ell} u_{q\ell} + \beta_{dj} \right], \quad \sum_{q,\ell} w_{dj,q\ell} = 1 \text{ when normalization is claimed.} \quad (\text{SCI-RTC-EQ-007})$$

The scale a is domain-qualified: equation (SCI-RTC-EQ-002) is the authorized raw `flxscale` route. Any other transfer carries separate identity and validity and does not import downstream calibration into RTC.

For a time-invariant FIR stage,

$$v_{d,j} = \sum_{\ell \in \mathcal{L}_k} h_{d,\ell}^{(k)} u_{d,j-\ell}, \quad H_d^{(k)}(e^{i\omega}) = \sum_{\ell \in \mathcal{L}_k} h_{d,\ell}^{(k)} e^{-i\omega\ell}, \quad H_d^{(k)}(1) = \sum_{\ell} h_{d,\ell}^{(k)}. \quad (\text{SCI-RTC-EQ-008})$$

For a causal IIR or notch stage, a state-complete form is

$$\mathbf{s}_{j+1} = A_k \mathbf{s}_j + B_k u_j, \quad v_j = C_k \mathbf{s}_j + D_k u_j, \quad \mathbf{s}_{j_0} = \mathbf{s}_{0,k}. \quad (\text{SCI-RTC-EQ-009})$$

An exactly equivalent recurrence is admissible only with its coefficient convention, precision, state, and boundary identity. A causal-stability claim requires every realized pole strictly inside the unit circle.

One possible selected mask operator is normalized masked FIR convolution,

$$v_j = \frac{\sum_{\ell} h_{\ell} m_{j-\ell} u_{j-\ell}}{\sum_{\ell} h_{\ell} m_{j-\ell}}, \quad \left| \sum_{\ell} h_{\ell} m_{j-\ell} \right| > \epsilon_m, \quad (\text{SCI-RTC-EQ-010})$$

but equation (SCI-RTC-EQ-010) is not a selected default. Segmenting, zero filling, restoring a mask after filtering, holding state, and renormalizing are distinct operators.

C.3 Phase-zero sampling and response

For a coherent pre-sampling segment of length N , v0.1 selects only

$$y_{d,n}^{\text{RTC}} = v_{d,Mn}^{\text{preD}}, \quad 0 \leq Mn < N, \quad f_{\text{out}} = \frac{f_{\text{in}}}{M}, \quad N_y = \begin{cases} 0, & N = 0, \\ 1 + \left\lfloor \frac{N-1}{M} \right\rfloor, & N > 0. \end{cases} \quad (\text{SCI-RTC-EQ-011})$$

The representative assigned-grid time is the selected point time $t_n^{\text{out}} = t_{Mn}$; the scientific support is nevertheless the complete transitive support $\mathcal{S}_{dn}$ of all preceding stages. A selected edge/output policy may reject candidate points, but shall not relabel a different phase or silently change equation (SCI-RTC-EQ-011).

For output (d, n) and acquired input (q, j) , the exact local response is

$$K_{dn,qj} = \frac{\partial y_{dn}}{\partial x_{qj}^{\text{acq}}} \quad \text{at fixed realized state } \Omega. \quad (\text{SCI-RTC-EQ-012})$$

At a selection boundary the derivative may be discontinuous or undefined; that state shall be represented rather than replaced by an LTI claim.

In the fixed, detector-separable, translation-invariant, single-rate interior case only,

$$H_d^{\text{comb}}(e^{i\omega}) = \prod_{k=1}^K H_d^{(k)}(e^{i\omega}), \quad \tau_g(\omega) = -\Delta t \frac{d}{d\omega} \text{unwrap}[\arg H_d^{\text{comb}}(e^{i\omega})]. \quad (\text{SCI-RTC-EQ-013})$$

Amplitude is $|H|$, power response is $|H|^2$, and phase is $\arg H$ under the recorded Fourier convention. Group delay is unavailable at response zeros unless a bounded convention is supplied.

For an LTI prefilter followed by phase-zero selection, the exact multirate identity is

$$Y(e^{i\Omega_o}) = \frac{1}{M} \sum_{r=0}^{M-1} H(e^{i(\Omega_o+2\pi r)/M}) X(e^{i(\Omega_o+2\pi r)/M}), \quad (\text{SCI-RTC-EQ-014})$$

under the declared Fourier/sign convention. Every folded image contributes; point selection itself is not anti-alias filtering.

For an FIR output, direct support starts with

$$\mathcal{S}_{dn}^{(1)} = \{Mn - \ell : \ell \in \mathcal{L}\}, \quad \mathcal{S}_{dn} = \text{closure}_{A,B,F,\text{state}}(\mathcal{S}_{dn}^{(1)}). \quad (\text{SCI-RTC-EQ-015})$$

IIR support extends to the declared state/reset or context boundary and may be mathematically infinite. Any finite guard is an approximation with a stated residual-tail criterion.

C.4 Moments, influence, and learned-plan selection

For admitted input mean $\boldsymbol{\mu}_x$ and full covariance Σ_x , conditional on fixed Ω independent of the modeled random noise,

$$\mathbb{E}[\mathbf{y} \mid \Omega] = L_\Omega \boldsymbol{\mu}_x + \mathbf{b}'_\Omega, \quad (\text{SCI-RTC-EQ-016a})$$

$$\Sigma_{y|\Omega}^{\text{stat}} = L_\Omega \Sigma_x L_\Omega^\top. \quad (\text{SCI-RTC-EQ-016b})$$

If selectors depend on the random data, unconditional covariance obeys

$$\text{Cov}(\mathbf{y}) = \mathbb{E}_\Omega[\text{Cov}(\mathbf{y} \mid \Omega)] + \text{Cov}_\Omega(\mathbb{E}[\mathbf{y} \mid \Omega]). \quad (\text{SCI-RTC-EQ-017})$$

The second term is not zero by default. With named nuisance parameters $\boldsymbol{\eta}$ and an adequate local linearization,

$$\Sigma_y^{\text{sys}} \approx J_\eta \Sigma_\eta J_\eta^\top, \quad J_\eta = \left. \frac{\partial \mathbb{E}[\mathbf{y} \mid \Omega, \boldsymbol{\eta}]}{\partial \boldsymbol{\eta}} \right|_{\boldsymbol{\eta}_0}, \quad (\text{SCI-RTC-EQ-018})$$

and a total-covariance claim requires every applicable component and cross term:

$$\Sigma_y^{\text{total}} = \Sigma_y^{\text{stat}} + \Sigma_y^{\text{sys}} + \Sigma_y^{\text{selection}} + \Sigma_y^{\text{model}} + \Sigma_y^{\text{cross}}. \quad (\text{SCI-RTC-EQ-019})$$

Unknown components are unavailable, not zero.

Define cause-preserving influence, direct exclusion, and consumer eligibility by

$$\mathcal{I}_{dn} = \text{closure}_{K, \text{generative links}}\{\text{typed source causes on } \mathcal{S}_{dn}\}, \quad (\text{SCI-RTC-EQ-020a})$$

$$E_{dn}^{\text{RTC,direct}} = V_d^{\text{det}} \wedge V_{dn}^{\text{num}} \wedge V_{dn}^{\text{support}} \wedge V_{dn}^{\text{response}} \wedge \neg D_{dn}^{\text{ALIGN_synth}} \wedge \neg D_{dn}^{\text{RTC_replace}},$$

$$E_{dn}^{(c)} = E_{dn}^{\text{RTC,direct}} \wedge P_c(\mathcal{I}_{dn}, \mathcal{S}_{dn}, K_{dn}, \dots) \quad (\text{SCI-RTC-EQ-020b})$$

D_{dn} identifies a directly selected synthesized or replaced occurrence. P_c is the declared policy of consumer c for noncenter transitive influence. RTC preserves every cause and numerical dependency but does not replace P_c with a universal transitive-ineligibility decision.

For learned mode, let $\mathcal{M}$ be the admitted finite integer candidate set and $\mathcal{F}(M)$ the conjunction of all owner-resolved astronomical transfer, exact phase-zero alias, sampling, realized-filter realizability, and downstream-compatibility constraints, evaluated with the smallest admitted beam and maximum valid in-scan speed. Then

$$\mathcal{M}_{\text{safe}} = \{M \in \mathcal{M} : \mathcal{F}(M) = \text{true}\}, \quad M_{\star} = \max \mathcal{M}_{\text{safe}}. \quad (\text{SCI-RTC-EQ-021})$$

Equation (SCI-RTC-EQ-021) is defined only after every numerical tolerance, support rule, fallback, and compatibility predicate named in the owner ledger is resolved. Percentile speed is diagnostic, not safety authority.

Finally, the atomic product is

$$\mathcal{O}_{\text{RTC},\mathcal{R}} = (\mathbf{y}, \mathcal{J}, \mathcal{K}, \mathcal{S}, \mathcal{I}, \mathcal{F}, \mathcal{V}, \mathcal{U}, \mathcal{P}, \mathcal{D}, \mathcal{C}), \quad (\text{SCI-RTC-EQ-022})$$

where $\mathcal{J}$ is ordered identity, $\mathcal{K}$ complete response status, $\mathcal{S}$ support, $\mathcal{I}$ influence, $\mathcal{F}$ typed flags/causes, $\mathcal{V}$ separated validity and eligibility inputs, $\mathcal{U}$ uncertainty/covariance availability, $\mathcal{P}$ one-way state/provenance, $\mathcal{D}$ diagnostics, and $\mathcal{C}$ atomic completion or required-output failure state.

C.5 Scientific design and learn–resolve–apply identities

For a locally Gaussian effective beam with angular covariance Σ and angular scan-velocity vector $\mathbf{v}$, the projected source-crossing time scale is

$$\sigma_t = (\mathbf{v}^T \Sigma^{-1} \mathbf{v})^{-1/2}. \quad (\text{SCI-RTC-EQ-023})$$

Equation (SCI-RTC-EQ-023) is a local Gaussian diagnostic. Scientific admission shall use the exact source-times-realized-response model over the declared beam, scan, source-class, and cadence domain.

A generic realized notch may be represented, where applicable, by

$$H_{\text{notch}}(z) = g \frac{\prod_{a=1}^{N_z} (1 - z_a z^{-1})}{\prod_{b=1}^{N_p} (1 - p_b z^{-1})}, \quad |p_b| < 1 \text{ for a causal-stability claim.} \quad (\text{SCI-RTC-EQ-024})$$

This form imposes no universal biquad. Exact realized coefficients, sample rate, frequency convention, phase/direction, state, edges, and precision are part of the response.

For an admitted FIR design family H_N , a selected order shall follow a declared constrained problem of the form

$$N_{\star} = \min_N \left\{ N : \begin{array}{l} \sup_{f \in \mathcal{P}} |H_N(f) - H_{\text{target}}(f)| \leq \epsilon_{\mathcal{P}}, \\ \sup_{f \in \mathcal{S}} |H_N(f)| \leq \epsilon_{\mathcal{S}}, \\ \text{phase/delay, alias, edge/guard, precision, and realizability predicates pass} \end{array} \right\}. \quad (\text{SCI-RTC-EQ-025})$$

The sets, metrics, tolerances, equality conventions, and family are owner-resolved; an approximate order formula is not universal authority.

For a causal linear-phase FIR with N taps at rate f_{samp} , its nominal interior group delay is

$$\tau_g = \frac{N-1}{2f_{\text{samp}}}. \quad (\text{SCI-RTC-EQ-026})$$

Where a temporal response has effective delay τ during a locally uniform scan, the corresponding scan-direction angular displacement is

$$\Delta \boldsymbol{\theta} = \mathbf{v} \tau. \quad (\text{SCI-RTC-EQ-027})$$

For every learned RTC operation, the lifecycle is

$$\Lambda = \text{Learn}(\mathcal{L}; \Theta^{\text{eff}}), \quad \Pi = \text{Resolve}(\Lambda, \Theta^{\text{policy}}), \quad \mathbf{y} = \mathcal{A}_{\Pi}(\mathbf{x}), \quad \frac{\partial \Pi}{\partial j} = 0 \text{ during apply.} \quad (\text{SCI-RTC-EQ-028})$$

The final equality denotes plan immutability over apply slots; it is not an assumption that the realized operator is stationary at masks, edges, state boundaries, or other declared exceptions.

For bounded iterative notch-plan refinement with original admitted input $x^{(0)}$, initial fixed or identity plan Π_0 , attempt index a , and accepted-plan index k_a available at the start of attempt a ,

$$\begin{aligned}
x^{\text{eval},(0)} &= \mathcal{A}_{\Pi_0}(x^{(0)}), & k_1 &= 0, \\
\Lambda_a &= \text{Learn}(\mathcal{L}_a, x^{\text{eval},(k_a)}, \Pi_{k_a}), \\
(\tilde{\Pi}_a, \delta_a) &= \text{Resolve}(\Lambda_a, \Pi_{k_a}, \Theta^{\text{policy}}), \\
k_{a+1} &= k_a + \mathbb{I}[\delta_a = \text{accepted}], \\
\delta_a = \text{accepted} &\implies \begin{cases} \Pi_{k_a+1} = \tilde{\Pi}_a, \\ x^{\text{eval},(k_a+1)} = \mathcal{A}_{\Pi_{k_a+1}}(x^{(0)}) \end{cases}, & (\text{SCI-RTC-EQ-029}) \\
\delta_a \neq \text{accepted} &\implies \Pi_{k_a+1} = \Pi_{k_a} \text{ and no new evaluation is required,} \\
x^{\text{final}} &= \mathcal{A}_{\Pi_K}(x^{(0)}), & K = k_{N_A+1} \leq N_A < \infty, \\
\frac{\partial \Pi_k}{\partial j} &= 0 \quad \text{during every apply of accepted plan } k.
\end{aligned}$$

Equation (SCI-RTC-EQ-029) separates finite attempts from the accepted-plan sequence, defines the initial evaluation for nonidentity Π_0 , and states the default original-input replay rule. A sequential cascade from a preceding evaluation product requires separate explicit operator authority and complete propagation of every intermediate parent, response, state, phase, edge, support, covariance, and uncertainty.

D Bounded assumptions and non-goals

SCI-RTC-ASM-001 – Capability.

V0.1 treats Stokes-I primary **xs**. Enabled polarimetry and measured R-channel execution are excluded until separately approved.

SCI-RTC-ASM-002 – Aligned-grid boundary.

The input grid, cadence, phase, mapping, gaps, synthesis/origin state, support, and scan partition are admitted ALIGN facts. Physical event timing, absolute timing accuracy, and a producer epoch may remain unavailable.

SCI-RTC-ASM-003 – Coordinate boundary.

Coordinate-dependent operations assume an admitted AST identity, frame, topology, detector binding, support, validity, and uncertainty where required. The absence of such authority does not license inference or clamping.

SCI-RTC-ASM-004 – Calibration boundary.

Calibration-factor production, atmosphere lineage, beam meaning, uncertainty, and promotion belong to SCI-BEAM/SCI-CAL. RTC remains in raw $\Delta f/f$ through replacement, filtering, and sampling. SCI-CAL may consume the exact RTC bundle and apply absolute **flxscale** plus target atmosphere only after RTC.

SCI-RTC-ASM-005 – Selected-policy boundary.

Despite, donor, mask, FIR, IIR, notch, state, edge, non-finite, recovery, sampling, diagnostic, and failure choices are selected versioned policies. No current default or numerical threshold is promoted to a universal scientific constant by this contract.

SCI-RTC-ASM-006 – Conditional linearity.

Linear response and covariance identities condition on fixed realized selection and state. Data-derived selectors make the unconditional operation generally nonlinear and nonstationary.

SCI-RTC-ASM-007 – Statistical input.

Moments or covariance are claimed only from a supplied statistical model with declared correlation and limitations. Absence does not imply diagonal, white, stationary, independent, or zero covariance.

SCI-RTC-ASM-008 – Learned-mode scope.

Every learned-then-fixed RTC operation uses one declared learning population, one resolved immutable plan, and one explicit compatibility domain. The first learned sampling scope produces one common observation plan. Per-array, per-scan, heterogeneous-cadence, noise-aware, and continuously adaptive sampling execution are successor scopes. Bounded iterative notch-plan refinement is admitted only as a finite outer sequence of attempts over complete immutable accepted plans; rejected attempts do not advance the accepted-plan sequence, and no parameter evolves within an apply pass.

SCI-RTC-ASM-009 – Analytical authority.

Exact analytical source-times-realized-filter transfer and phase-zero alias accounting define learned-plan admis-

sion. Deterministic vectors may falsify a realization, while source injections and observational studies are required for separately claimed performance or science-impact evidence.

SCI-RTC-ASM-010 – Compact representation.

Dense observation-sized response, covariance, or per-sample provenance is not required when a factorized, sparse, interval-based, or generative representation exactly reconstructs the operator, support, causes, and exceptions.

SCI-RTC-ASM-011 – Downstream authority.

RTC supplies causal inputs but does not own PTC cleaning/weights, VAL eligibility policy, MAP estimation, BEAM inference, empirical NOI products, MAP-003 response tracing, map filtering, source fitting, or fruit-loop recurrence.

SCI-RTC-ASM-012 – Claim separation.

The contract is an algebraic and semantic authority only. It contains no claim that a present implementation conforms, that a representation is faithful, that observational performance is adequate, or that a mode is ready for production.

D.1 Explicit non-goals

This revision does not derive ALIGN timing, AST coordinates, SCI-BEAM or SCI-CAL calibration, PTC, VAL, MAP, or FLT estimators, source fitting, or recurrence; authorize absolute-timing or astrometric corrections; redesign mature despiking/FIR/IIR/notch numerics; choose universal numerical thresholds; authorize online adaptation; activate polarimetry or measured R; prescribe source classes, files, storage schemas, or product names; authorize per-array/per-scan cadence; inspect implementation; perform validation or reductions; or assert production status.

E Sequential normative requirements

The following identifiers are stable within v0.1/r0.4. Conformance is requirement-by-requirement; satisfying a neighboring requirement does not waive another one.

Area	ID	Normative statement
Boundary	SCI-RTC-REQ-001	RTC shall admit only an aligned primary xs stream whose signal is raw, uncalibrated detector $\Delta f/f$ and whose ordered identity, assigned grid, support, origin/synthesis state, and validity are declared.
Boundary	SCI-RTC-REQ-002	Every requested RTC product role shall declare sign, reference/baseline, valid domain, plan, and required consumers while retaining the raw $\Delta f/f$ signal identity; no role shall give xs a calibrated meaning.
Boundary	SCI-RTC-REQ-003	Every RTC conditioned detector output, including Beammap, Pointing, OOF, Science, and diagnostic-only roles, shall remain raw $\Delta f/f$ and shall not be silently replaced by a calibrated timestream or map.
Boundary	SCI-RTC-REQ-004	A calibrated detector signal shall be a distinct downstream SCI-CAL product in the top-of-atmosphere, point-source-equivalent mJy-per-fixed-nominal-beam convention, with admitted factor, atmosphere, detector, validity, uncertainty, and provenance state.
Boundary	SCI-RTC-REQ-005	RTC shall preserve the exact raw parent, plan, response, support, factor authority, and atmosphere lineage required by the downstream CAL handoff but shall not apply, derive, repair, promote, or reinterpret SCI-CAL calibration.
Identity	SCI-RTC-REQ-006	Observation, Tune, array, network/interface, detector occurrence, APT binding, native row, aligned slot, scan, science interval, context interval, output slot, stage, and product identity shall remain distinct and immutable.
Identity	SCI-RTC-REQ-007	The ordered detector and sample shapes, index bases, native-to-aligned mapping, aligned-to-output map, scan membership, and memory order shall be explicit; indices shall not serve as cross-artifact detector identity.
Identity	SCI-RTC-REQ-008	RTC shall preserve the admitted ALIGN cadence, phase, rate, gaps, support, and synthesis/origin state without claiming physical integration-event timing or absolute epoch authority.
Identity	SCI-RTC-REQ-009	A coordinate-dependent operation shall consume the exact AST identity, frame, topology, detector binding, support, and validity required by its product role, with no implicit frame conversion.
State	SCI-RTC-REQ-010	Requested, effective, observation-resolved, learned/resolved when applicable, and realized state shall be recorded separately in one-way order; later state shall not overwrite the accepted request.
State	SCI-RTC-REQ-011	Outer, inner, full, mini, diagnostic, simulated, processed, and any other produced stage roles shall have distinct immutable identities and explicit parent/processing links without requiring duplicate computation.
Policy	SCI-RTC-REQ-012	Every despike, donor, replacement, mask, filter, state, edge, non-finite, recovery, sampling, flag, diagnostic, and fallback action shall be governed by an admitted versioned selected policy rather than an undocumented default.
Order	SCI-RTC-REQ-013	RTC shall complete raw-domain alignment, compatible donor transfer/replacement, temporal filtering, and phase-zero sampling before SCI-CAL applies absolute flxscale and target-atmosphere correction as in equation (SCI-RTC-EQ-004); no calibrated RTC branch or alternative interleaving is authorized in v0.1.

Area	ID	Normative statement
Donor	SCI-RTC-REQ-014	A raw donor q to raw target d using the detector-static comparison coordinate $\chi_i = \phi_i x_i$ shall use ϕ_q/ϕ_d only when both flxscale factors are valid for the exact occurrences under the same compatible convention/domain and $\phi_d \neq 0$.
Donor	SCI-RTC-REQ-015	The raw donor route in REQ-014 shall use flxscale and shall neither require nor restore a scientific role for legacy responsivity .
Donor	SCI-RTC-REQ-016	When REQ-014 is not available, replacement shall fail under that route unless a separate donor-to-target authority declares direction, quantity/unit domain, support, uncertainty, validity, and provenance.
Donor	SCI-RTC-REQ-017	The despite detector, target indicator, candidate eligibility, topology, score, tie rule, donor weights, time relation, source-mask/flag use, fallback, and every realized donor link shall be reconstructible.
Donor	SCI-RTC-REQ-018	A replaced target shall retain target identity, link every donor and factor, preserve the replacement cause, and expose correlations from reused or shared donors; finiteness shall not create independent exposure.
Influence	SCI-RTC-REQ-019	An output whose directly selected occurrence is ALIGN-synthesized shall be universally excluded as an independent detector measurement; noncenter synthesized influence shall preserve its numerical effect, typed cause, and support for the downstream consumer's declared eligibility policy.
Influence	SCI-RTC-REQ-020	An output whose directly selected occurrence is RTC-replaced shall be universally excluded as an independent detector measurement; noncenter replacement influence shall preserve its numerical effect, typed cause, donor linkage, and support for the downstream consumer's declared eligibility policy.
Filter	SCI-RTC-REQ-021	Every enabled response-changing stage shall record chronological order, exact full-precision coefficients or exact digest, coefficient convention, normalization, numeric precision, rate, direction, state, context, mask, edge, non-finite policy, and exceptions once per coherent segment.
Filter	SCI-RTC-REQ-022	An FIR claim shall expose exact impulse support, DC gain, phase convention, and normalization; a label such as "normalized" shall not substitute for the coefficient sum or response.
Filter	SCI-RTC-REQ-023	An IIR or notch claim shall expose an exact recurrence or state-space equivalent, realized poles/zeros where applicable, initial/final state, reset/carry boundary, transient/guard rule, precision, and stability status.
Mask	SCI-RTC-REQ-024	A processing mask shall be represented as an operator control, including geometry or selection, boundary convention, dilation, denominator/segment behavior, and state interaction; it shall not be treated as acquisition validity or confidence.
Mask	SCI-RTC-REQ-025	Invalid, unavailable, wrong-frame, out-of-domain, or ambiguously bound coordinates shall not become outside-source samples and shall fail the affected required coordinate operation.
Finite	SCI-RTC-REQ-026	A non-finite required sample, factor, coefficient, coordinate, or filter state shall be rejected, marked unavailable over its exact transitive influence, or handled by an explicitly authorized prior replacement/recovery rule; it shall not be silently coerced to zero.
Edge	SCI-RTC-REQ-027	Boundary extension, incomplete context, filter guards, IIR recovery, short/empty scans, and output selection shall be explicit; an edge or incomplete-state cell shall not be labeled an ordinary interior sample.

Area	ID	Normative statement
Sampling	SCI-RTC-REQ-028	V0.1 downsampling shall use only phase-zero point selection $y^{\text{RTC}}[n] = v^{\text{preD}}[Mn]$ from the final pre-decimation stream; arithmetic-mean and other block-aggregate downsampling are excluded.
Sampling	SCI-RTC-REQ-029	Each coherent segment shall record M , zero phase, input/output rates and grids, the cardinality of equation (SCI-RTC-EQ-011), selected-point representative time, full transitive support, edge disposition, and flag/influence aggregation.
Sampling	SCI-RTC-REQ-030	Any alias-free or bounded-alias claim shall include the exact realized prefilter and all folded bands of equation (SCI-RTC-EQ-014), with a declared input band, metric, bound, and unavailable state; point selection alone shall not be called anti-alias filtering.
Sampling	SCI-RTC-REQ-031	Fixed mode shall resolve an admitted factor/filter/rate/compatibility plan before application and record it independently of realized output.
Learned	SCI-RTC-REQ-032	Every learned RTC operation shall preserve distinct requested, effective, observation-resolved, learned-evidence, resolved-plan, and applied states and shall not advertise a plan before admitted evidence and numeric policy are complete.
Learned	SCI-RTC-REQ-033	Learned resolution shall choose the largest admitted integer factor only after astronomical-transfer, exact phase-zero alias, sampling, filter-realizability, and downstream-compatibility constraints all pass using the smallest admitted beam and maximum valid in-scan speed.
Learned	SCI-RTC-REQ-034	The first learned applied scope shall use one common immutable observation cadence and filter across arrays and scans; percentile speed shall remain diagnostic rather than safety authority.
Learned	SCI-RTC-REQ-035	Apply shall consume exactly one immutable resolved plan and shall not retune, adapt, or silently fall back; an unavailable or rejected plan shall follow the selected failure/fallback policy before apply begins.
Learned	SCI-RTC-REQ-036	Restart shall restore a state-complete resolved plan, verify exact plan/input compatibility, and reject stale or incompatible beam, speed, scan, coefficient, rate, phase, role, or parent identity.
Response	SCI-RTC-REQ-037	The atomic bundle shall contain the complete realized response of SCI-RTC-DEF-010 or a typed unavailable state; a partial filter kernel shall not be labeled the conditioned response.
Response	SCI-RTC-REQ-038	Selection, detector mixing, masking, finite boundaries, state, or multirate behavior shall use a local/factorized detector-time response or truthful unavailability rather than an unjustified scalar transfer; downstream SCI-CAL response remains a separate composition.
Response	SCI-RTC-REQ-039	A scalar LTI response shall be claimed only on a stated uniform, fixed-state, detector-separable, translation-invariant interior domain with no data-derived selector or undeclared rate change.
Response	SCI-RTC-REQ-040	Where defined, response shall distinguish amplitude, power, phase, group delay, sampling rate, Fourier convention, alias images, support, and response uncertainty; undefined phase/delay at a response zero shall remain unavailable.
Response	SCI-RTC-REQ-041	Output support shall be expanded transitively through alignment, replacement, every filter, state boundary, edge rule, and phase-zero selection; a selected center index shall not stand in for full support.

Area	ID	Normative statement
Statistics	SCI-RTC-REQ-042	Conditional mean and covariance shall follow equations (SCI-RTC-EQ-016a)–(SCI-RTC-EQ-016b) only for an admitted input model and fixed realized state, with reused donors and overlapping support represented as correlations.
Statistics	SCI-RTC-REQ-043	When selection depends on modeled random data, selection bias and selection covariance shall be modeled or explicitly unavailable; they shall not be set to zero merely by conditioning on the realized selector.
Statistics	SCI-RTC-REQ-044	Calibration, atmosphere, beam, donor-scale, coefficient, cadence/phase, response, and model nuisance terms shall retain value, validity, uncertainty, provenance, and correlation scope, including required cross terms.
Statistics	SCI-RTC-REQ-045	Unknown covariance shall remain unavailable and shall not be replaced by diagonal, white, stationary, independent, zero, or scalar-weight assumptions; conditional inverse variance shall not be called full precision or total uncertainty.
Validity	SCI-RTC-REQ-046	Acquisition support, original validity, ALIGN synthesis, numerical computability, coordinate validity, flags/causes, operator masks, replacement, transitive influence, complete support, response status, consumer eligibility, uncertainty availability, and provenance shall remain separate states.
Validity	SCI-RTC-REQ-047	A flag shall describe a cause and shall not automatically become a mask, invalidity, weight, or universal eligibility decision; aggregation shall preserve every contributing cause required by the selected policy.
Output	SCI-RTC-REQ-048	RTC shall produce the role-specific atomic bundle of equation (SCI-RTC-EQ-022); conditioned TOD alone shall not be reported as a complete required RTC product.
Output	SCI-RTC-REQ-049	Required malformed identity/state or missing authority shall propagate to atomic required-output failure; optional unavailable provenance detail shall be typed and numerically inert.
Output	SCI-RTC-REQ-050	Compact provenance shall reconstruct input/parent identity, detector binding, grid and intervals, downstream CAL handoff authority, donor links, ordered filters, masks, coefficients/state/edges, sampling, response/uncertainty status, exceptions, counts, and all one-way state records.
Output	SCI-RTC-REQ-051	Each diagnostic shall name estimator, unit, support, mask/selection, validity, availability, and whether it is inert, advisory, or enters the selected policy; optional diagnostic detail shall not change values, flags, eligibility, cardinality, or failure.
Consumer	SCI-RTC-REQ-052	Beyond the direct exclusions in REQ-019–020, RTC shall supply cause-preserving transitive influence rather than a universal downstream eligibility bit; each consumer shall bind the exact RTC parent, declare its influence-to-eligibility policy, and shall not erase causes or strengthen unavailable timing, coordinate, calibration, response, covariance, or eligibility claims.
Consumer	SCI-RTC-REQ-053	PTC-disabled terminal RTC output shall be a valid requested state when selected; no PTC or map product shall be implied, while any requested downstream product shall require the exact admitted RTC bundle.
Claims	SCI-RTC-REQ-054	Algebraic contract correctness, implementation conformity, representation fidelity, observational performance, science-impact qualification, and production readiness shall be stated and assessed as separate claims.

Area	ID	Normative statement
Lifecycle	SCI-RTC-REQ-055	Every learned operation shall name its learning population, estimator, support, selection, exclusions, quality, uncertainty, validity, and compatibility domain; same-observation learning shall be identified as data-dependent.
Lifecycle	SCI-RTC-REQ-056	Resolve shall convert evidence and fixed policy into one immutable plan containing every applicable sequence, family, coefficient identity, rate/convention, order/taps, band/notch parameter, applicability, mask, donor, edge/guard/reset, decimation/timing, response/support, uncertainty, validity, and downstream-compatibility field.
Lifecycle	SCI-RTC-REQ-057	A plan shall be resolved only when every required predicate and tolerance passes; missing evidence, missing tolerance, rejected evidence, or unavailable compatibility shall not be treated as pass.
Lifecycle	SCI-RTC-REQ-058	Any coefficient or operator parameter that evolves during apply shall be governed by a separately authorized online-adaptive estimator contract; it shall not be represented as learned-then-fixed filtering.
Design	SCI-RTC-REQ-059	Every despoke, donor, mask, notch, low/high/band-pass, FIR, IIR, decimation, edge/state, or non-finite operation shall declare scientific purpose, signal/contaminant model, mathematical operator, design parameters, learn/resolve/apply behavior, astronomical and calibration consequences, validation/falsification, and unavailable states.
Design	SCI-RTC-REQ-060	Filter admission shall use the exact source-times-realized-response over the declared scan-speed/direction, projected-beam, source-class, prior-response, and cadence domain; $f \sim v/\theta$, a scalar FWHM, or a nominal cutoff shall remain pedagogical only.
Notch	SCI-RTC-REQ-061	Every notch shall resolve center, rate/frequency convention, width or Q, depth, family/order, poles/zeros where applicable, phase/group delay, state, transient/guard, applicability, learned uncertainty, drift tolerance, and combined response for overlaps.
Bands	SCI-RTC-REQ-062	Every admitted low-, high-, or band-pass shall resolve pass/transition/stop bands, ripple, attenuation, phase/delay, family, order/taps, precision, edges/guards, retained astronomical modes, and unavailable modes; no class label shall substitute for exact response.

Area	ID	Normative statement
FIR	SCI-RTC-REQ-063	FIR order or tap count shall follow owner-resolved transfer, alias, phase, support, guard, precision, and realizability constraints such as equation (SCI-RTC-EQ-025); an arbitrary integer or approximate family formula shall not be scientific authority.
Donor	SCI-RTC-REQ-064	Donor replacement shall be identified as a continuity surrogate at the target identity and shall separately establish numerical convention transfer, donor admissibility, and scientific eligibility; compatible <code>flxscale</code> shall not be evidence of common sky signal.
Donor	SCI-RTC-REQ-065	Beammap-role donor scaling shall use an explicitly identified prior APT or other pre-existing calibration authority and shall not use a new <code>flxscale</code> derived from the same Beammap conditioning lineage.
Sampling	SCI-RTC-REQ-066	Decimation factor and prefilter shall be justified from cadence, exact response, folded-band alias bounds, smallest projected crossing time, maximum valid scan speed, source class, timing/coordinate accuracy, and downstream compatibility.
Timing	SCI-RTC-REQ-067	Each plan shall resolve temporal registration through zero phase, shifted output time, coordinate correction, response-aware downstream use, or an approved bound on residual centroid/PSF effect; selected grid time alone shall not imply response centering.
CAL	SCI-RTC-REQ-068	Calibration transfer between Beammap and target processing shall bind the complete RTC plans and require matching response or explicit calculation, renormalization, or observational evidence; identical operation labels or preserved units are insufficient.
Validation	SCI-RTC-REQ-069	Validation of a notch, broad-band/FIR, donor, or decimation plan shall vary its scientific design parameters over the admitted scan/beam/source domain and measure astronomical transfer, phase/centroid, PSF/morphology, ringing, support/edge loss, covariance, and alias or interference residual as applicable.
Claims	SCI-RTC-REQ-070	No plan shall be called science-qualified solely because coefficient algebra, optimization, serialization, or deterministic operator fixtures pass.

Area	ID	Normative statement
Iteration	SCI-RTC-REQ-071	Iterative notch-plan refinement shall use a finite owner-approved attempt bound and stopping policy; attempt count and accepted-plan count shall remain distinct, and an unbounded repeat-until-satisfactory process is unavailable.
Iteration	SCI-RTC-REQ-072	Every apply of accepted plan Π_k shall consume exactly that immutable plan and shall not change coefficients, notch parameters, policy, masks, donor rules, decimation, or state conventions.
Plan	SCI-RTC-REQ-073	Every proposed successor $\tilde{\Pi}_a$ shall be a complete cumulative plan that dispositions all prior and proposed notches and records all stages, order, coefficients/precision, state, masks, edges/guards, intervals, decimation/timing, response/support, uncertainty, validity, and compatibility; rejection shall not assign it an accepted-plan index.
Replay	SCI-RTC-REQ-074	The default evaluation and final production rule shall apply each complete accepted plan to original admitted $x^{(0)}$ as in equation (SCI-RTC-EQ-029); exploratory attempts shall not accumulate hidden rounding, padding, state, duplicate-filter, mask, flag, or parent effects.
Cascade	SCI-RTC-REQ-075	A cascade from a preceding filtered product shall require explicit scientific-operator authority and immutable intermediate parents, with total response, state, phase, edges, support, covariance, uncertainty, and sequential parentage retained.
Evaluate	SCI-RTC-REQ-076	Each successor attempt shall learn from the current accepted-plan evaluation and assess intended line attenuation and residual, parameter error/drift, predicted-versus-measured response, stability, ringing/transient, edge/guard loss, phase/delay, source amplitude/centroid/PSF, extended transfer, covariance, and support where applicable.

Area	ID	Normative statement
Candidate	SCI-RTC-REQ-077	A successor line shall be admitted only after comparison of original, predicted, and conditioned spectra, cumulative response, line uncertainty, astronomical spectrum, transfer/alias budgets, and detector/network/array/scan/interval consistency excludes notch-edge, ringing, transient, leakage, normalization, numerical, and response-floor artifacts.
Budget	SCI-RTC-REQ-078	Successor admission shall preserve owner-approved cumulative interference improvement, removed bandwidth, astronomical transfer, source amplitude/centroid/PSF, phase/registration, ringing/transient, guard/support, stability, cadence/response compatibility, uncertainty, and provenance budgets.
Stability	SCI-RTC-REQ-079	Plan stability shall compare the complete line set and matching identity, centers, widths/Q, depths, family/order, precision, applicability, state/reset, masks/guards, broad-band stages, decimation/timing, response/support, and compatibility over the required consecutive accepted-plan comparisons.
Stop	SCI-RTC-REQ-080	Every attempt shall record one typed accepted, rejected, or stop disposition plus its cause; maximum-attempt, oscillating-plan, and nonconvergent-parameter terminations shall not be labeled convergence or plan stability, and rejection shall leave the last accepted plan available unless policy says otherwise.
Lineage	SCI-RTC-REQ-081	Every attempt shall preserve attempt index, current accepted-plan index, original input, previous accepted plan, learning parent/population and PSD definition, original/predicted/measured spectra, proposal and disposition, predicates/tolerances, any newly accepted complete plan/evaluation product, cumulative response/astronomical diagnostics, stop cause, and final plan identity in one-way state.
Restart	SCI-RTC-REQ-082	Serialization and restart shall reproduce the final selected complete plan and declared parent rule exactly and reject incompatible original input, rate, learning population, parent, coefficient, interval, state, or policy identity.

F Falsifiable predictions and limiting cases

Each prediction is an algebraic or semantic falsification target. A future conformance activity may choose an exact numerical tolerance justified by the operation and representation before observing candidate output. No result is reported by this document.

ID	Construction and prediction	Falsifier
SCI-RTC-PRED-001	Identity chain: raw role, $A = I$, no replacement or filter, $M = 1$, valid original samples. Values, raw $\Delta f/f$ unit, order, identities, cardinality, response, flags, and support are unchanged.	Any hidden calibration, value/order/cardinality change, flag loss, nonidentity kernel, or new claim of timing/coordinate authority.
SCI-RTC-PRED-002	Role separation: feed the same raw xs input to Beammap, Pointing, OOF, Science, and diagnostic-only RTC roles. Every RTC output remains $\Delta f/f$; a distinct calibrated output exists only after downstream SCI-CAL.	Any role gives xs a calibrated meaning, an RTC product is labeled mJy/beam, or CAL output is silently interchanged with RTC output.
SCI-RTC-PRED-003	Raw donor direction: choose $\phi_q = 2$, $\phi_d = 4$, and donor $x_q = 3$. Equation (SCI-RTC-EQ-002) predicts raw target replacement 1.5 and equal detector-static comparison coordinates $\chi_q = \chi_d^R = 6$.	Use of ϕ_d/ϕ_q , output other than 1.5, unequal comparison coordinate, calibration claimed from χ , or dependence on legacy responsivity .
SCI-RTC-PRED-004	Raw donor availability: independently make either factor invalid, convention/domain-incompatible, occurrence-mismatched, or set $\phi_d = 0$. The flxscale route is unavailable.	Finite replacement, stale factor reuse, row-number-only binding, zero insertion, or success without separate donor authority.
SCI-RTC-PRED-005	Order: trace a raw input through replacement, every temporal stage, and phase-zero sampling, then through distinct downstream SCI-CAL. The parent and response follow equation (SCI-RTC-EQ-004).	Absolute calibration or target atmosphere appears inside RTC, flxscale donor transfer is mislabeled calibration, or the CAL parent omits the complete RTC plan.
SCI-RTC-PRED-006	Interior target/donor impulses under fixed selection. Each output column equals equation (SCI-RTC-EQ-012); donor energy appears in the declared detector column, support, scale, and phase.	Missing or circularly wrapped energy, unexplained mixing, wrong donor identity/direction, support, normalization, or phase.
SCI-RTC-PRED-007	Constant input with sufficient context. Output equals the constant times realized DC gain after the declared transient; a DC-preserving chain is exactly constant.	Drift, divide-by-zero mask normalization, edge values advertised as interior, or a false DC-preservation claim.
SCI-RTC-PRED-008	Interior step with complete context. Response equals the cumulative impulse response and settles according to DC gain and stability.	Undeclared overshoot, failure to settle, hidden state leakage, or disagreement with the impulse-derived response.
SCI-RTC-PRED-009	Affine ramp through each FIR and admitted alignment mapping. Output slope and offset follow the zeroth and first moments of the exact kernel.	Unexplained clipping, wrong phase/slope/offset, or a ramp-preservation claim without the required moments.
SCI-RTC-PRED-010	Long bin-centered and off-bin sinusoids at DC, passband interior/edge, transition, stopband, and near output Nyquist. Amplitude and phase follow equations (SCI-RTC-EQ-013) and (SCI-RTC-EQ-014).	Gain/phase outside preregistered tolerance, unrecorded frequency convention, or an aliased component labeled passband signal.
SCI-RTC-PRED-011	Sinusoids at the realized notch center and symmetric offsets under the exact rate and state. Depth, width, phase, and transient follow realized zeros/poles.	Center/depth/width mismatch, unstable pole, rate mismatch, hidden state, or an edge transient reported ordinary-valid.
SCI-RTC-PRED-012	Coordinate mask with valid inside/outside values plus invalid, unavailable, wrong-frame, out-of-domain, and ambiguously bound coordinates. Invalid classes fail the affected operation and remain distinct from outside.	Invalid becoming outside/unmasked, entering geometry, clamping to a center/edge, or loss of the distinct cause.
SCI-RTC-PRED-013	Samples just inside, exactly on, and just outside each selected mask boundary and temporal dilation. Membership follows the recorded inclusive/exclusive and dilation policy identically across execution modes.	An undocumented boundary, frame mismatch, asymmetric dilation, or sequence-dependent membership.

ID	Construction and prediction	Falsifier
SCI-RTC-PRED-014	Impulses and constants at first/last support cells and immediately inside complete FIR context. Values, guard, support, and output admission follow the selected edge rule.	Undeclared padding, truncation, wrap, renormalization, guard, or output selection.
SCI-RTC-PRED-015	Zero, impulse, and constant inputs split and re-joined at scan/chunk/state boundaries for an IIR. Results follow the declared reset/carry and terminal-state policy.	Prior-observation leakage, hidden carry/reset, unbounded non-finite poisoning, or contradiction between split and declared state semantics.
SCI-RTC-PRED-016	Lengths $0, 1, L - 1, L, L + 1$ around every support/guard threshold, with stable scan identity.	Crash, underflow, stale output, silent scan deletion/renumbering, or ordinary-valid status without required context.
SCI-RTC-PRED-017	Inject NaN, $+\infty$, and $-\infty$ separately into a sample, factor, coordinate, coefficient, and IIR state. Failure or unavailable influence follows REQ-026 exactly.	Silent zero/coercion, finite-looking output without cause, stale state recovery, or an unbounded tail reported valid.
SCI-RTC-PRED-018	Place synthesized and replaced causes first at the directly selected occurrence and then at noncenter members of full support. Direct cases are universally excluded; noncenter causes propagate unchanged and consumer eligibility follows the declared policy.	A direct case called independent, a noncenter cause erased or made universally ineligible by RTC, or consumer disposition reported without its policy.
SCI-RTC-PRED-019	For $M = 1, 2, 3, 4, 5$ and $N = 0$ through several incomplete final strides, enumerate $j = Mn$. Cardinality and selected time follow equation (SCI-RTC-EQ-011); no block mean appears.	Wrong N_y , dropped/duplicated point, nonzero phase, averaged value, timestamp misidentification, or rate mismatch.
SCI-RTC-PRED-020	Two sinusoids separated by an integer multiple of f_{out} , before and after the realized pre-filter. Folded amplitude is the sum in equation (SCI-RTC-EQ-014).	A folded term missing from response, inadequate stopband labeled alias-free, or incorrect image frequency/amplitude.
SCI-RTC-PRED-021	Reuse one donor for two targets and give neighboring outputs overlapping filter support. Equation (SCI-RTC-EQ-016b) predicts cross-target and temporal covariance when the admitted input covariance permits it.	Independent weights assigned, off-diagonal terms silently dropped, or reused donor described as additional exposure.
SCI-RTC-PRED-022	Enable a response-changing mask, donor mix, stateful filter, edge, and decimator while providing only a scalar FIR transfer. Complete response state is unavailable.	The partial scalar transfer labeled complete, or downstream use that assumes omitted stages are identity.
SCI-RTC-PRED-023	Learned candidates include at least two safe factors and one failing each astronomical-transfer, alias, sampling, realizability, and compatibility constraint. Equation (SCI-RTC-EQ-021) chooses the largest passing factor.	A smaller safe factor selected without rule, any failing factor applied, a percentile speed used for safety, or a missing numeric predicate treated as pass.
SCI-RTC-PRED-024	Serialize and restart a learned plan, then alter one of beam identity, speed support, scan set, coefficient precision, rate, phase, role, or parent. Apply rejects the mismatch.	Silent relearning, retuning, native fallback, stale-plan application, or realized state rewriting the request.
SCI-RTC-PRED-025	Execute two observations sequentially with distinct rates, constants, factors, masks, donors, states, and output parents.	Any coefficient, state, factor, donor, mask, rate, flag, or product link leaks from the first observation.
SCI-RTC-PRED-026	Toggle optional detailed provenance and inert diagnostics while holding the selected policy and realized operator fixed. Values, flags, influence, eligibility, cardinality, response, and failures are identical.	Any scientific or numerical result changes, or an observational/production claim appears from the toggle.

ID	Construction and prediction	Falsifier
SCI-RTC-PRED-027	Bind identical learned parameters first to a separate training population and then to the apply observation. The numerical fixed-state response can match, but the second plan records data dependence and parameter/selection uncertainty.	Same-observation learning represented as fixed a priori, or conditional linearity used to erase selection or parameter uncertainty.
SCI-RTC-PRED-028	Remove in turn one required learning-population identity, tolerance, uncertainty, or downstream-compatibility predicate before resolve. The dependent plan remains unresolved and apply does not begin.	Missing evidence treated as pass, a default inserted, or output represented as applied learned mode.
SCI-RTC-PRED-029	Serialize one resolved plan and attempt during apply to alter a notch center, cutoff, tap count, donor rule, or decimation factor. The mutation is rejected or belongs to a separately authorized adaptive estimator.	Hidden retuning, silent fallback, or a realized record that rewrites the resolved plan.
SCI-RTC-PRED-030	Sweep notch center error, line drift, width, depth, Q/order, initialization, direction, and overlap. Residual line and astronomical amplitude, phase, centroid, covariance, transient, and edge loss follow the exact combined response.	Only line disappearance measured, or any changed astronomical observable omitted from the plan's validity evidence.
SCI-RTC-PRED-031	Scan the same projected source at multiple speeds and directions through a fixed filter. Its temporal spectrum and retained amplitude/PSF change according to equation (SCI-RTC-EQ-023) and exact response.	One nominal cutoff or scalar beam treated as speed- and direction-independent proof of preservation.
SCI-RTC-PRED-032	Design FIR candidates over multiple tap counts and coefficient precisions. The selected N_* is the smallest candidate passing every registered transfer, alias, phase, support, guard, and realizability predicate.	Arbitrary configured taps, approximate order formula promoted to authority, or a candidate with any failed predicate selected.

ID	Construction and prediction	Falsifier
SCI-RTC-PRED-033	Inject compact and extended sources through admitted high- or band-pass candidates. Low-frequency rejection produces the predicted lost broad modes, compensated response, or negative lobes.	Unit preservation labeled flux preservation, or removed absolute/extended modes reported available without response restoration evidence.
SCI-RTC-PRED-034	Choose donor and target with compatible <code>flxscale</code> but deliberately different sky signal or beam coupling. Raw-domain convention transfer remains algebraically correct while the surrogate is not a missing-sky reconstruction.	Compatible factors claimed to prove physical donor equivalence or independent exposure.
SCI-RTC-PRED-035	Attempt Beammap donor scaling with a new <code>flxscale</code> derived from that same conditioned Beammap. The route is rejected until an identified prior factor authority is supplied.	Circular use of the new factor or absence of the prior authority in lineage.
SCI-RTC-PRED-036	Apply filters with known nonzero group delay to Pointing/Beammap scans in opposite directions. Centroid displacement follows equation (SCI-RTC-EQ-027) with opposite scan-direction sign unless time/coordinates are corrected.	Selected grid time treated as response center, wrong displacement sign, or unrecorded correction.
SCI-RTC-PRED-037	Process calibrator and target with plans that share a nominal label but differ in coefficients, rate, phase, state, edge, or decimation. Cross-plan calibration transfer is unavailable absent explicit response accounting or observational evidence.	Nominal label, preserved unit, or <code>flxscale</code> alone treated as RTC-plan compatibility.
SCI-RTC-PRED-038	Pass all deterministic coefficient and operator fixtures but withhold the required source/scan design studies. Algebraic correctness may be supported; science qualification remains unassessed.	Deterministic pass promoted to observational performance, science qualification, or production readiness.

ID	Construction and prediction	Falsifier
SCI-RTC-PRED-039	Inject one strong coherent line and no other valid line. Attempt 1 accepts and suppresses it; Attempt 2 finds no acceptable successor. The accepted-plan count remains one and Π_1 is final with a no-successor stop.	A manufactured Π_2 or evaluation product, unbounded iteration, unsupported second notch, rejected evidence lost, or no typed stop/final-plan identity.
SCI-RTC-PRED-040	Inject two unequal coherent lines so the weaker becomes identifiable only after the stronger is suppressed. A later attempt accepts one complete two-notch plan as the next Π_k and replays it once on $x^{(0)}$; output follows the predicted cumulative response.	Attempt and accepted-plan indices conflated, increment-only plan, final use of an accumulated exploratory product, missing first notch, duplicate filtering, or response disagreement.
SCI-RTC-PRED-041	Create a peak at a first-notch edge that agrees with predicted cumulative ringing or response artifact. The successor candidate is rejected and the first plan remains final.	Artifact admitted as a line, response floor ignored, or prior accepted plan invalidated solely by rejection.
SCI-RTC-PRED-042	Compare applying a one-notch plan twice with applying the intended complete successor plan once to $x^{(0)}$. Values, response, phase, edges, support, and covariance differ unless exact operator equivalence is proved.	Repeated filtering treated as replay, or equality claimed from nominal notch identity without complete operator proof.

ID	Construction and prediction	Falsifier
SCI-RTC-PRED-043	Propose a successor that fails one cumulative transfer, stability, support, or compatibility predicate. The attempt is rejected; its evidence and cause persist while the accepted-plan index and final product remain unchanged.	Rejected proposal assigned a new accepted-plan index or evaluation, failed plan applied, prior product rewritten, rejected evidence erased, or missing predicate treated as pass.
SCI-RTC-PRED-044	Alternate between proposing addition and removal of the same line or vary its parameters outside declared stability across attempts. Termination is typed oscillating or nonconvergent.	Alternation or parameter instability labeled convergence, stable plan, or successful maximum-safe refinement.
SCI-RTC-PRED-045	Supply a sequence that reaches the owner-approved maximum attempt count without satisfying plan stability. The result records maximum-attempt termination and the last accepted plan, not convergence.	Maximum count labeled stable/converged, continued iteration, attempt count confused with accepted-plan count, or disappearance of the last accepted plan.
SCI-RTC-PRED-046	Serialize and restart the final cumulative plan, then vary original parent or learning-population identity. Exact compatible restart reproduces the product; either mismatch is rejected.	Silent relearning, stale-parent application, altered plan, or acceptance of incompatible learning lineage.

F.1 Analytic limits

The following limits are required: a compatible donor pair with $\phi_q = \phi_d$ gives unit raw donor scale; a replacement indicator tending to zero gives identity; filter coefficients tending to a Kronecker delta give identity; $M \rightarrow 1$ gives no rate change; a source-mask radius tending to zero selects only exact valid coincidences under the declared boundary; complete context approaches the declared interior response; and a causal pole approaching the unit circle exposes increasing memory and never retains a stability claim at or outside the circle.

End of shared normative core: SCI-RTC-v0.1/r0.4. View-specific material cannot add to or modify this scientific authority.